
ISSCC 2025 Tutorials

Fundamentals of DRAM I/O : Standards & Circuits

Hye-Ran Kim

Feb. 16, 2025

Outline

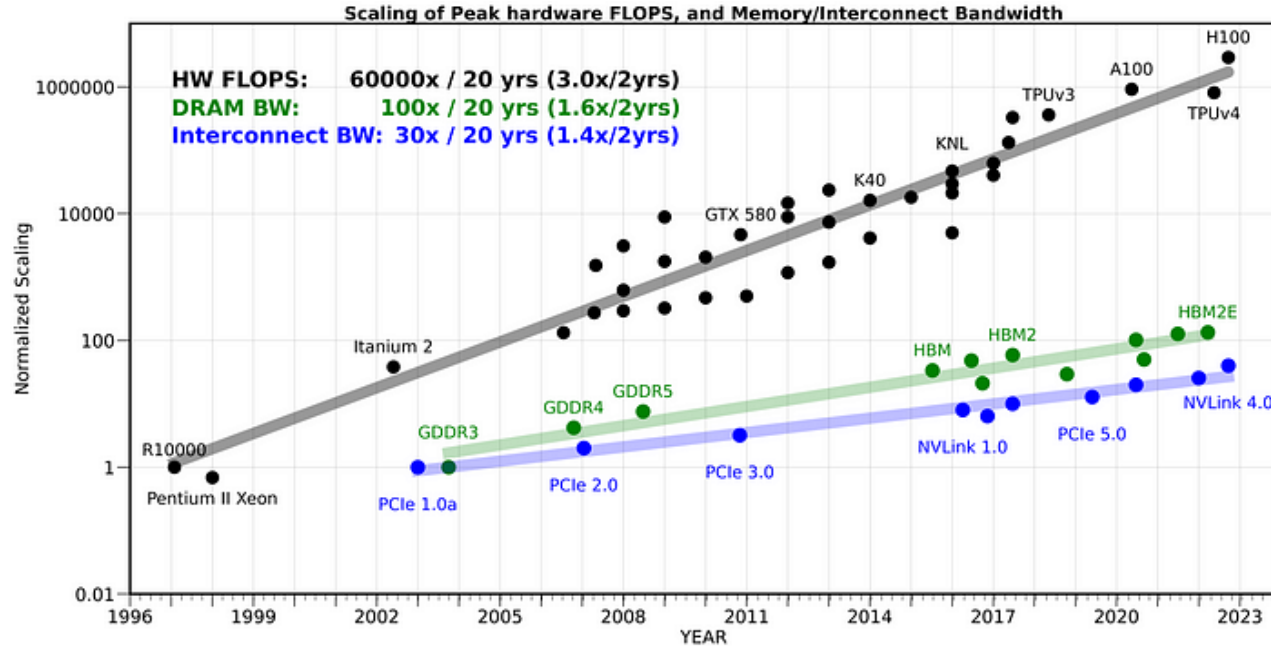
- DRAM I/O and Applications
 - DRAM Application
 - Signaling and Clocking
- I/O Speed Extension
 - Equalization
 - Training and Calibration
 - Link Protection
- Present and Future DRAM
- Conclusion

Self Introduction

- PhD degree from Sungkyunkwan University, Korea in 2020
- Have been with Samsung Electronics since 2006 and Principal Engineer (2019 – Now)
- Developed various commodity DRAMs
 - GDDR5/6/7 and LPDDR4/5
- Memory Standardization in JEDEC
 - GDDR7 and LPDDR5
- Member of ISSCC Technical Program Committee (2021 – Now)
- Research interests : I/O interfaces and Memory Architecture



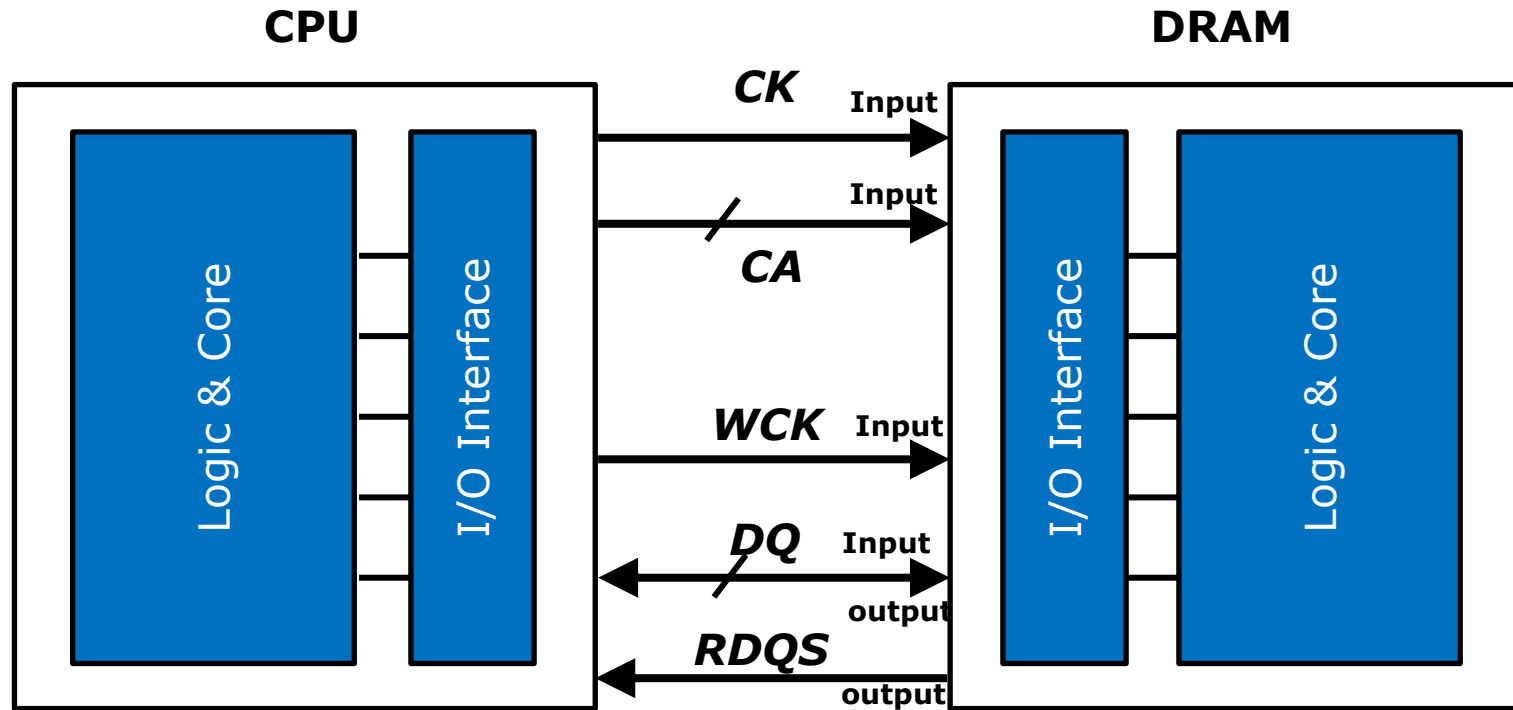
DRAM Trend in ISSCC2025



*<https://medium.com/riselab/ai-and-memory-wall-2cb4265cb0b8>

- ❑ Many core computing enabled the dramatic enhancement in GPU performance.
→ However, the pace of DRAM BW improvement is catching up only half of that.
- ❑ Not only bandwidth but also power efficiency improvement is important to satisfy the thermal constraint of the system.

What's Memory I/O?



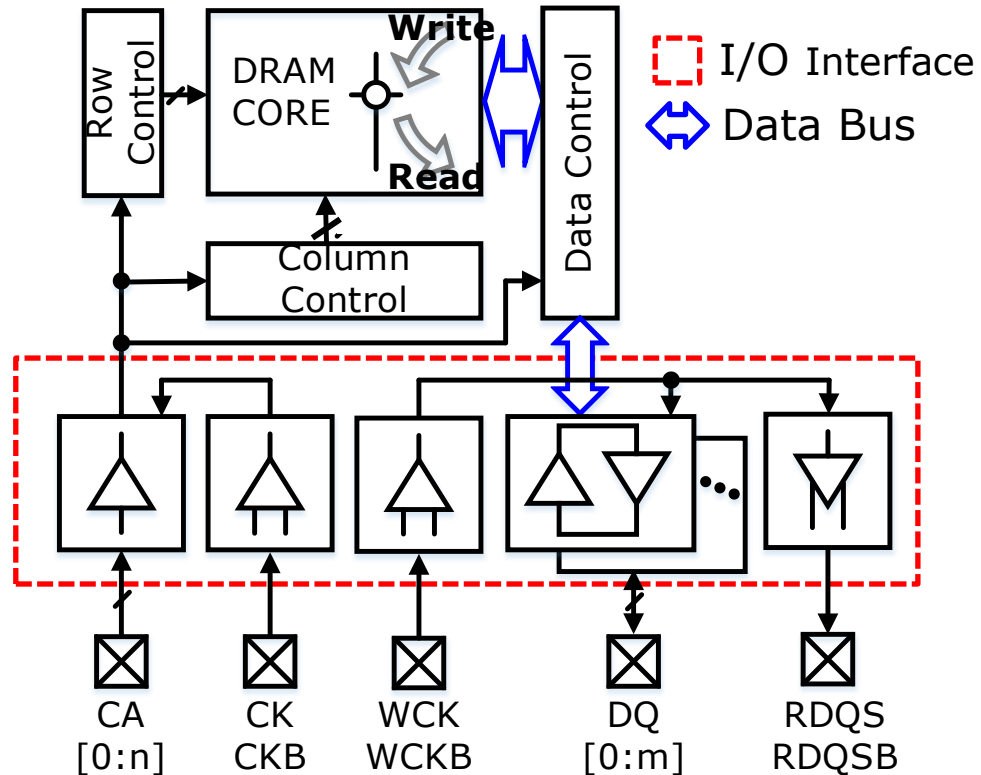
I/O = Input / Output

❖ Pin Description

- ✓ CK : Clock for command & address
- ✓ CA : Command and Address
- ✓ WCK : Clock for data
- ✓ DQ : Data transfer
- ✓ RDQS : Read strobe clock
- ** CA and DQ show parallel IO configurations.

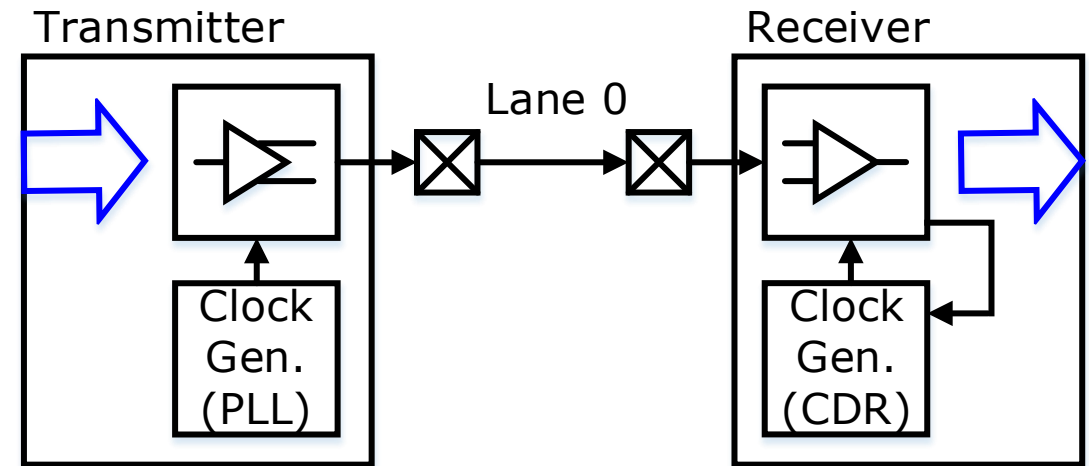
- ❑ Definition : Input/Output Pins and Relevant Circuitry (= I/O Interface)
- ❑ Purpose : To communicate with SoC (CPU, GPU, ...)

What's Memory I/O?



- ❑ Bi-directional DQ pins : Data in for Write, Data out for Read
- ❑ Differential clocks (CK, WCK, RDQS) vs. Single-ended data (DQ and CA)
- ❑ Multiple DQs (or CAs) are synchronized by WCK (or CK).

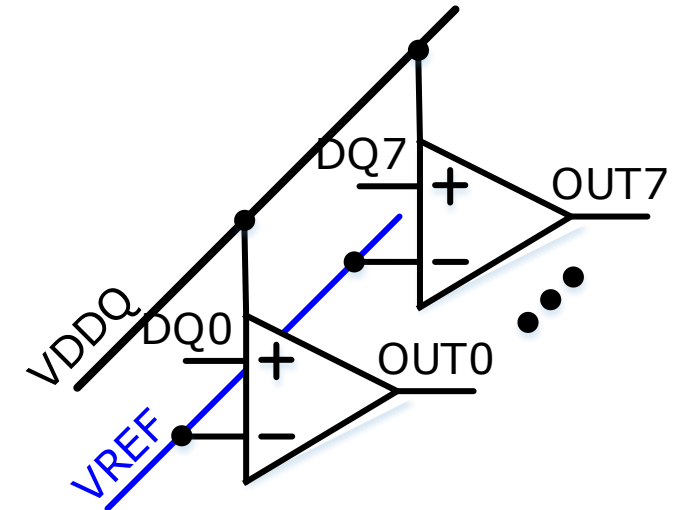
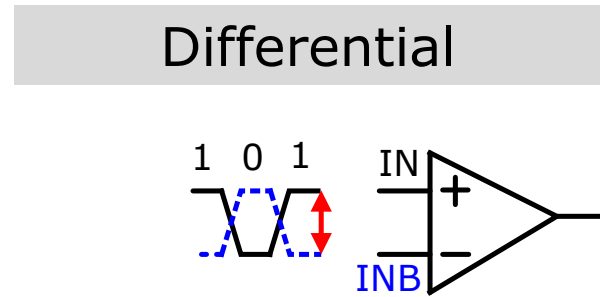
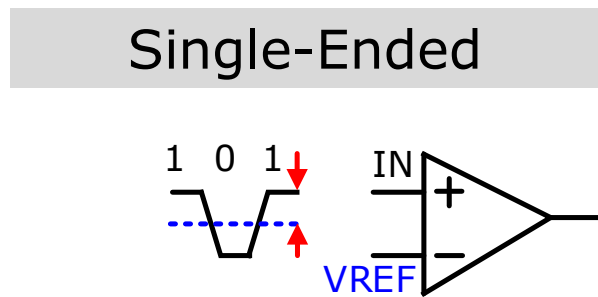
❖ Serial Link Example (e.g. PCIe)



- ✓ Lane : to transfer Command, Address, and Data
- ✓ Separate clock generators for a transmitter and a receiver : PLL (phase-locked loop) or CDR (clock-data recover)
- ✓ Usually differential signaling for high speed.

What's Memory I/O?

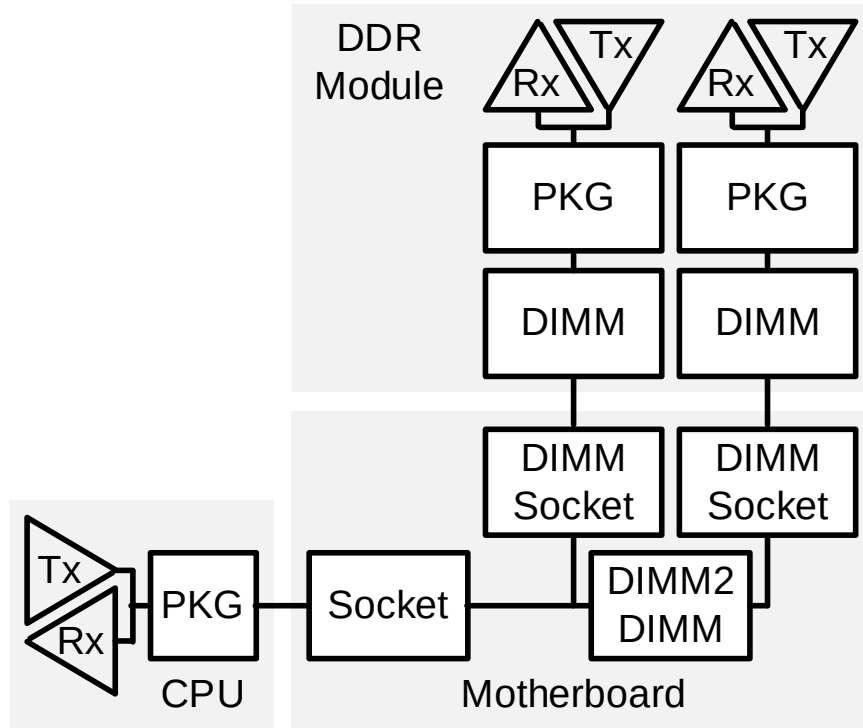
- DRAM IO Spec's are defined as **Single – Ended** signaling for cost reduction (pin count matters!).
** For CA and DQs except clock signals



❖ Consideration point :

- ✓ Pseudo differential input receiver : Less voltage margin than fully differential type.
- ✓ Vref Noise : Noise in Vref reduces voltage margin
- ✓ SSO / SSI (Simultaneous switching output/input) noise : Coding such as DBI-DC to minimize the simultaneous switching (Not shown in differential type!)
- ✓ Crosstalk : Each signal is required to be shielded in PCB for better SI (Signal Integrity).

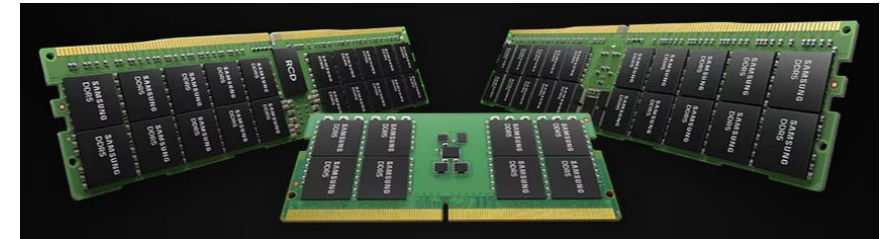
DRAM Applications : DDR



**** DIMM : Dual In-line Memory Module**



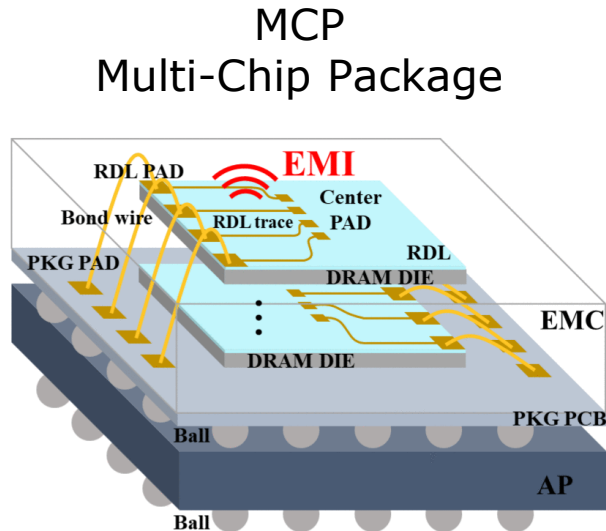
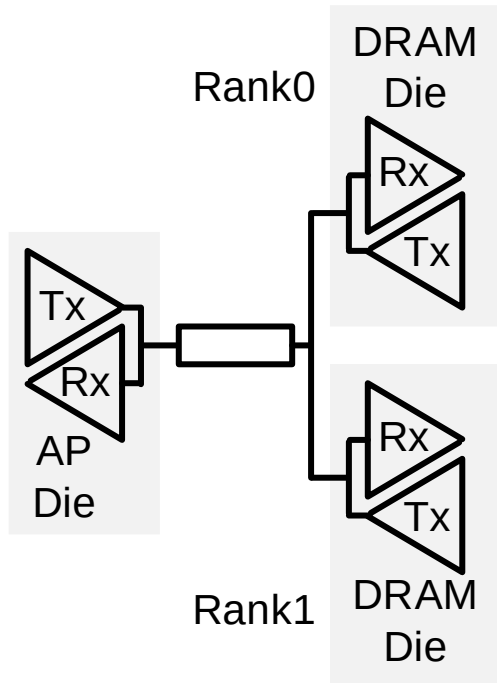
<https://www.intel.com>



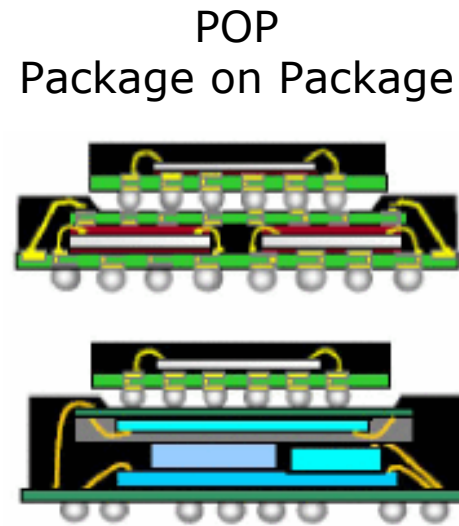
<https://semiconductor.samsung.com/dram>

- ❑ To obtain high density, Multi-DIMM required → 1 CPU + multi-DRAM
- ❑ Long channel and multi-drop → channel loss↑ and reflection↑
- ❑ Used in Server, PC (Desktop and Notebook), ...

DRAM Applications : LPDDR



J. H. Cho, S. Jeong, J. -B. Kim, J. D. Ihm
and J. Kim, doi:
10.1109/TEM.2023.3312680.

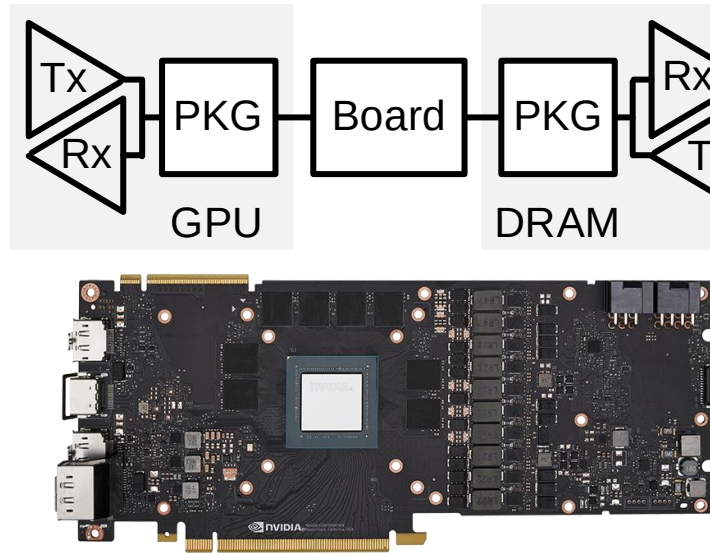


J. Y. Lee, J. Ahn, J. Yoo, J. Kim, H. -S. Park
and S. Okabe, doi:
10.1109/ICEPT.2006.359648.



- ❑ Multi-DRAM inside PKG to increase total density.
- ❑ Point-to-Point Channel (single or dual loads) → moderate channel loss
- ❑ Used in Notebook, Tablet, Smartphone, ...

DRAM Applications : GDDR



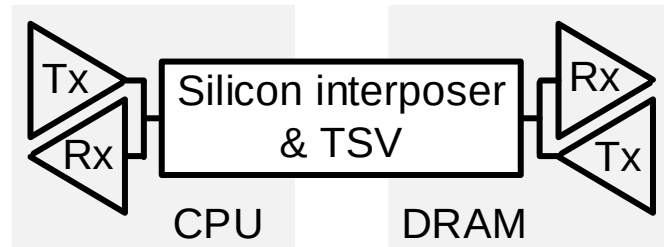
<https://www.playstation.com/>



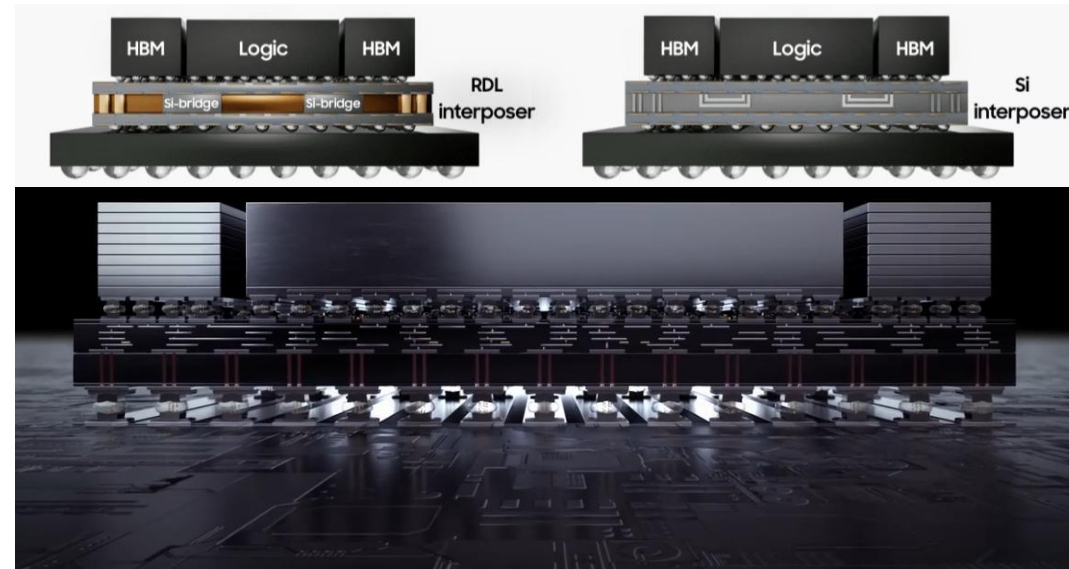
<https://www.nvidia.com/>

- ❑ Point-to-Point channel and only single load → Reduced Channel loss
- ❑ Highest speed among DRAMs
- ❑ Used in Graphic Card in PC, Gaming console, HPC, ...

DRAM Applications : HBM



* TSV = Through Silicon Vias



<https://semiconductor.samsung.com/foundry>

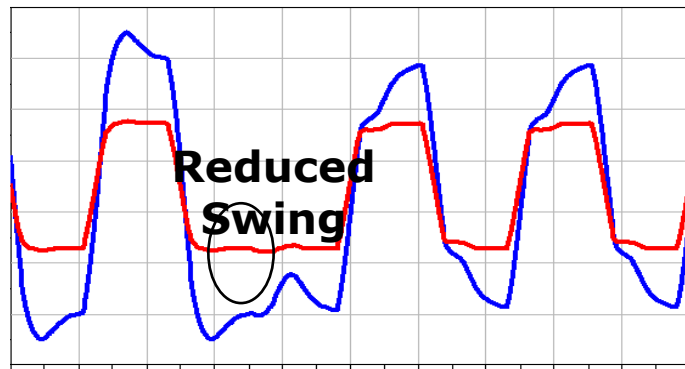
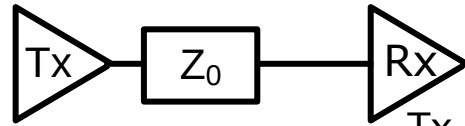
- ❑ Point-to-Point channel based on Silicon Interposer
- ❑ More than 1024 DQs for high bandwidth and performance
- ❑ Used in HPC for AI and ML, Server, ...

Outline

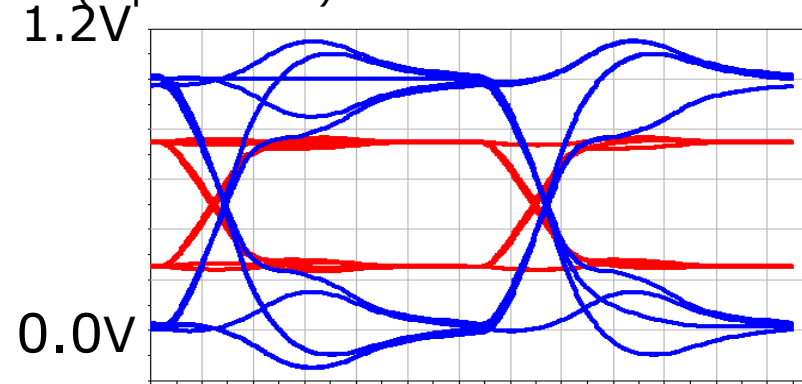
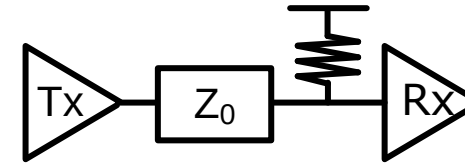
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 - **Signaling and Clocking**
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Why Termination?

Without Termination



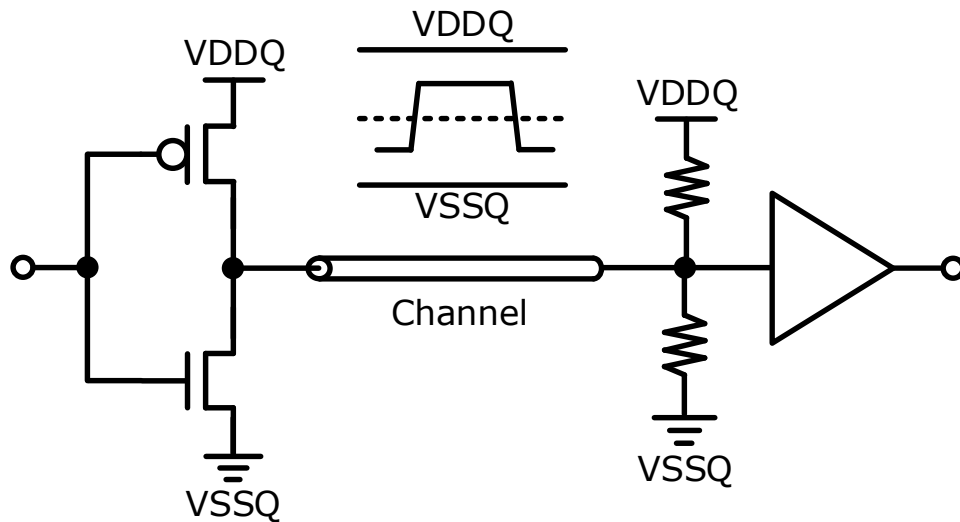
With Termination



- ❑ To improve signal quality, termination is necessary.
- ❑ Termination absorbs the signal energy and prevents the reflection effects.

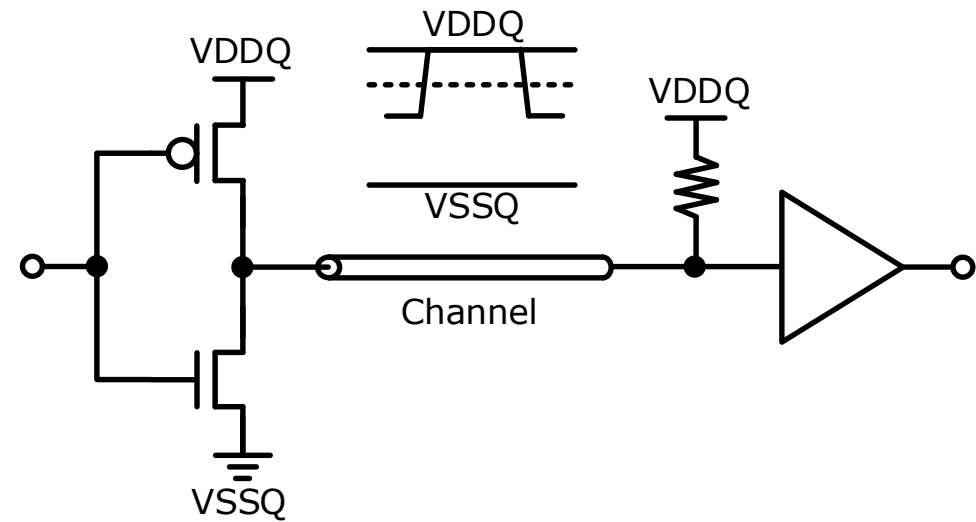
Signaling (1)

□ CTT (Center-Tap Termination)



- Small swing referenced to $\frac{1}{2}$ VDDQ level.
- Inappropriate for low power due to static current.

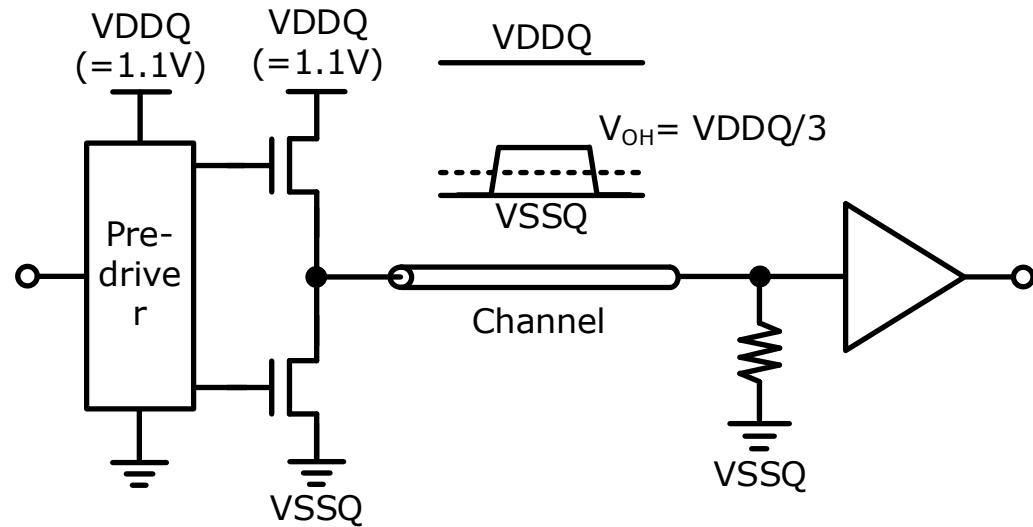
□ POD (Pseudo-Open Drain)



- Smaller than VDDQ-VSSQ (Rail-to-Rail) swing.
- Used in high-speed interfaces

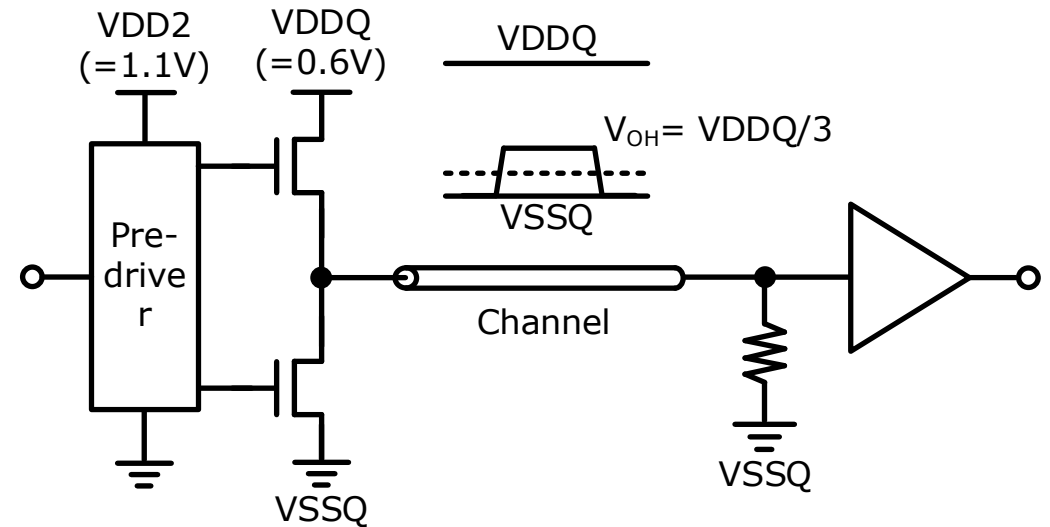
Signaling (2)

□ LVSTL (Low-Voltage Swing Terminated Logic)



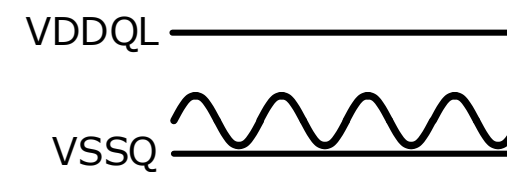
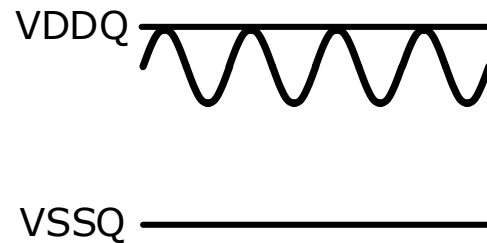
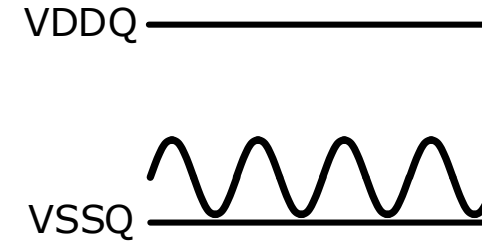
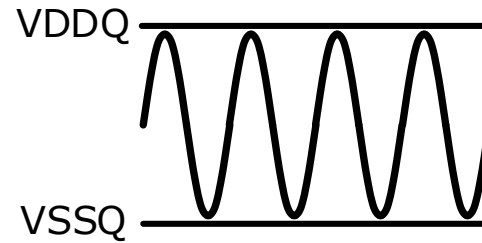
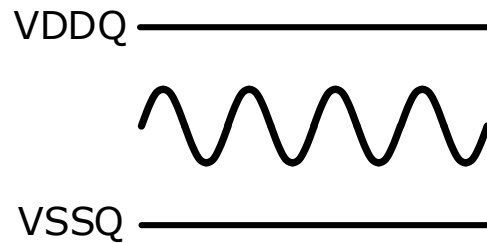
- Ground-reference signaling with small swing ($VDDQ/3$).
- C_{IO} (Input-capacitance) reduction due to NMOS-only implementation.

□ PI-LVSTL (Power-Isolated LVSTL)



- Main driver in VDDQ domain (much less than 1V).
- More power reduction through the small swing of $VDDQ/2$.

Signaling (3)



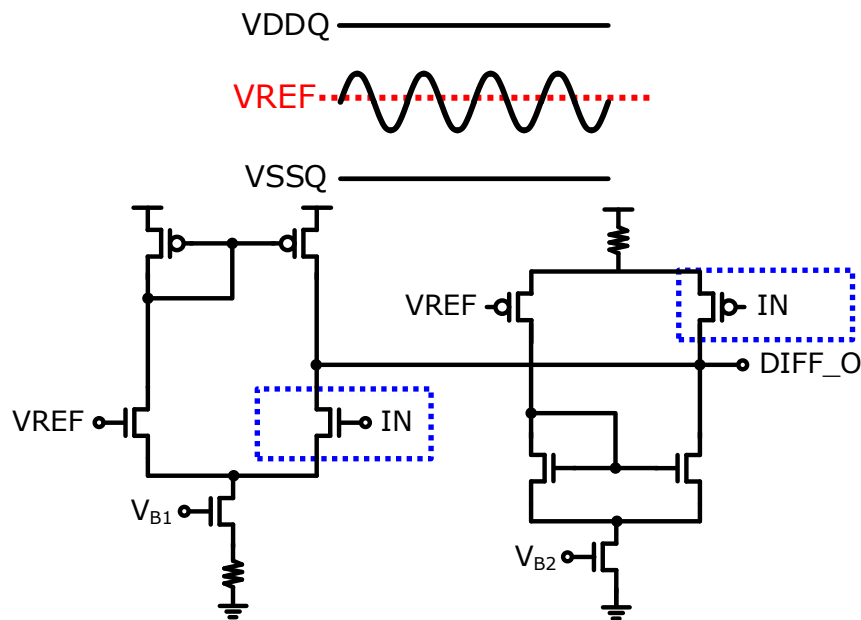
- Different level and swing according to each signaling
→ Different circuit implementations!

Input Buffer Comparison

- Signaling types determine input-buffer types regarding the input-detection capability.

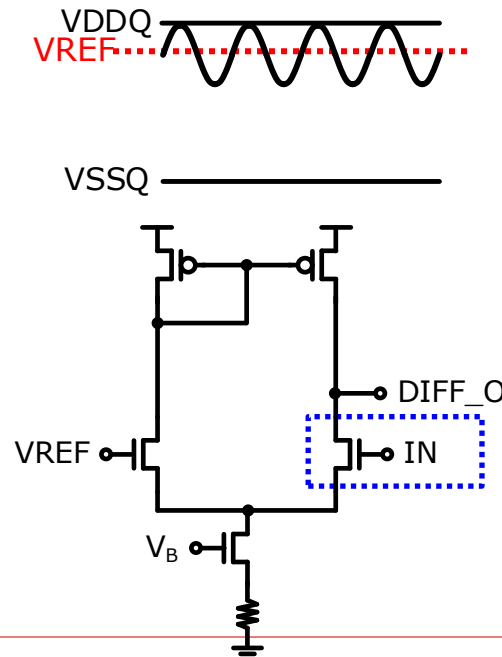
CTT

- Mid-level swing
→ N+PMOS input Tr.



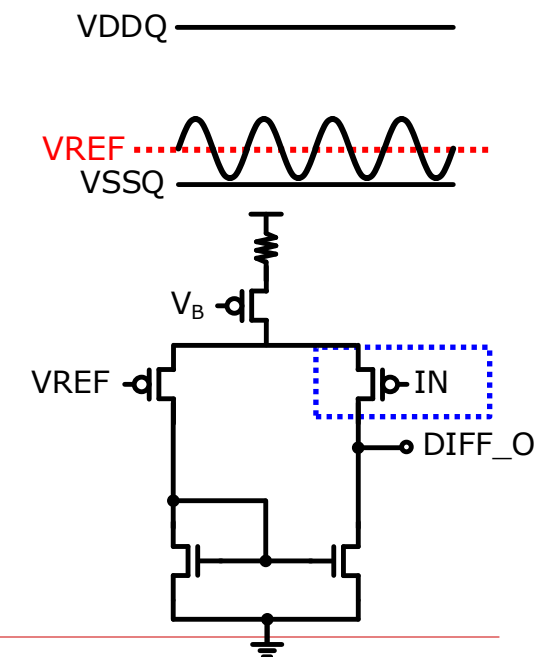
POD

- $VDDQ \sim VDDQ/2$ swing
→ NMOS input Tr.

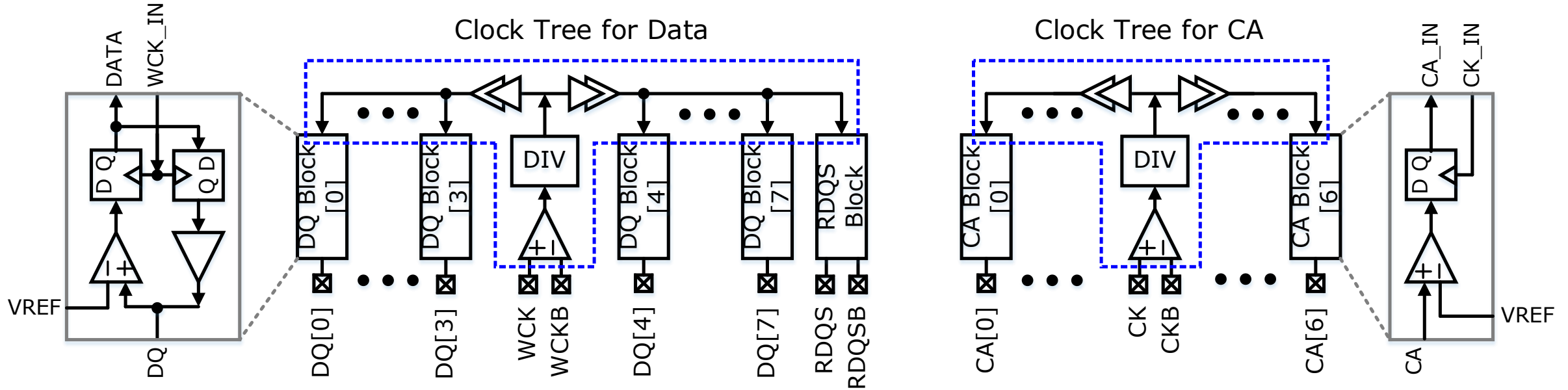


LVSTL (PI-LVSTL)

- $VDDQ/2 \sim VSS$ swing
→ PMOS input Tr.

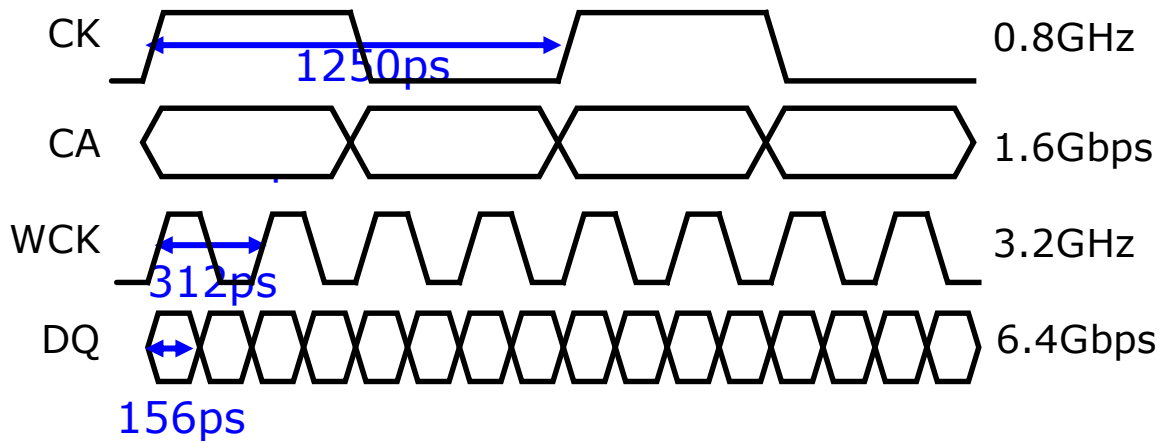


Clocking (1)

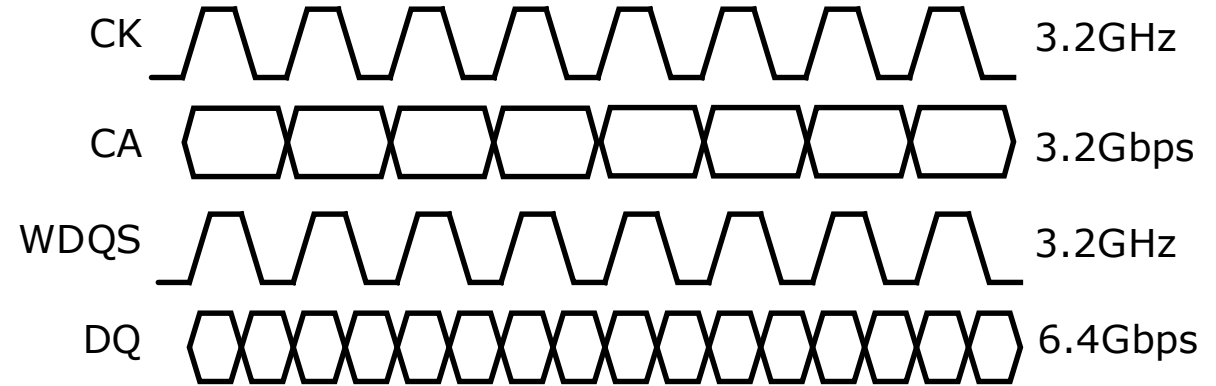


- ❑ Example of clock trees are shown for DATA and CA.
- ❑ Clock-tree design consideration :
 - 1) Circuit types (CML, CMOS, ...), 2) Number of Phases (1,2,4, ...)
- ❑ ***The quality of clock distribution mostly determines the maximum operating speed and the power consumption of I/O circuitry.***

Clocking (2)



LPDDR5X @ 6.4Gbps

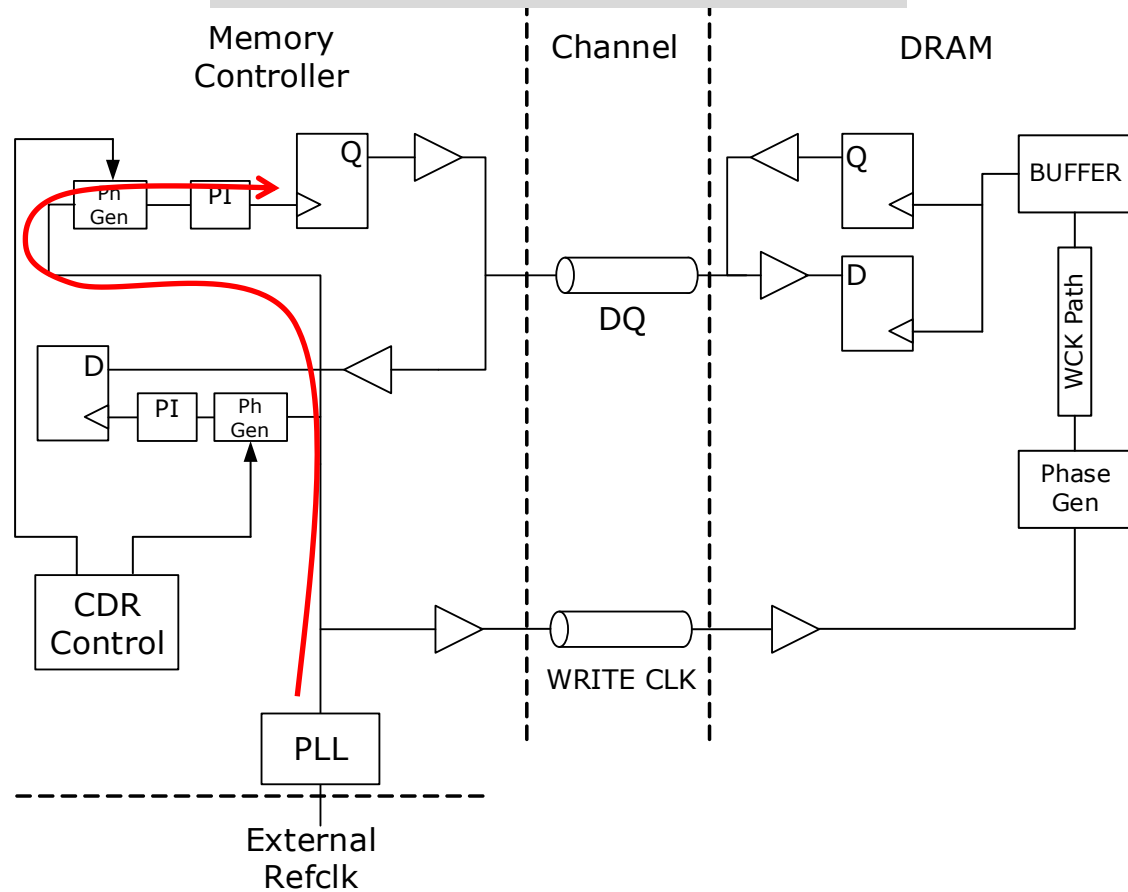


DDR5 @ 6.4Gbps

- ❑ WCK (WDQS) frequency depends on the maximum data-rate requirement.
- ❑ CK (CA) frequency and the ratio of CK to WCK (or WDQS) depends on the system.

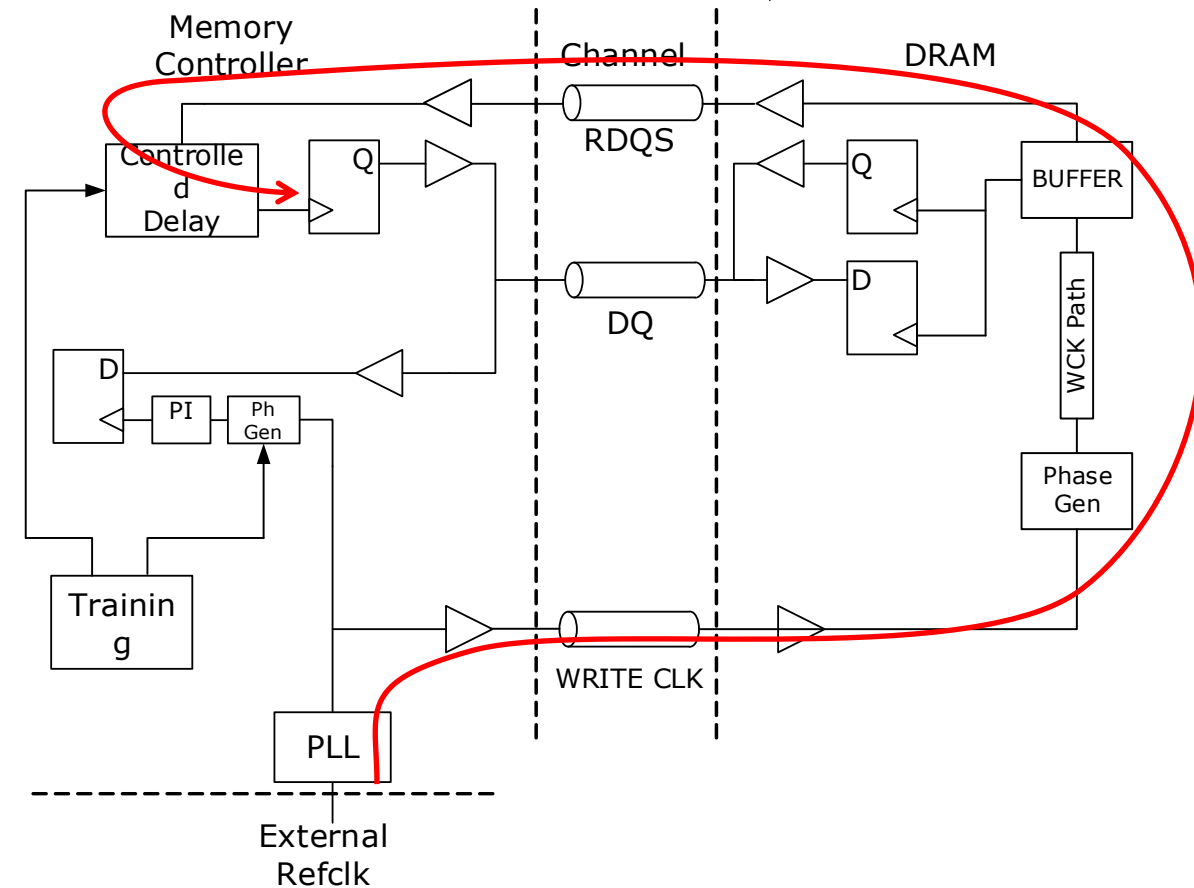
Clocking (3) : WCK vs. DQS

CDR based Clocking



- ✓ Pros : Pin Count & Power Reduction
- ✓ Cons : Vulnerable to Power Noise

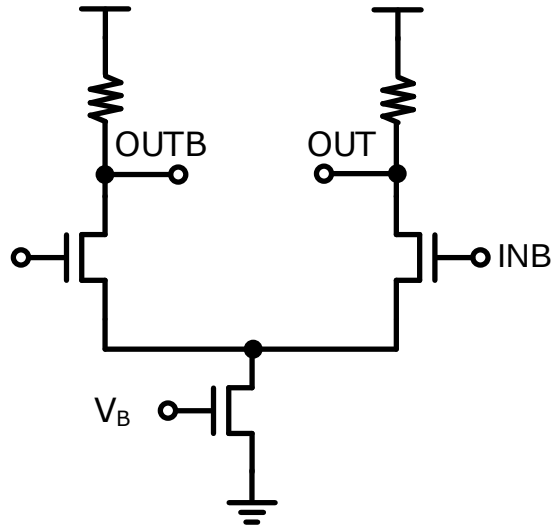
Source Synchronous Clocking



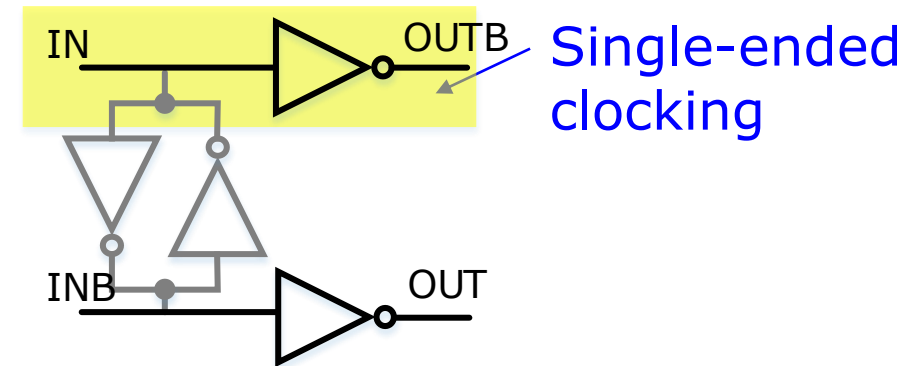
- ✓ Pros : Jitter Tracking between DQ & DQS (better IO margin)
- ✓ Cons : Pin Count & Power Increase

Clocking (4) : CML vs. CMOS

□ CML (Current-Model Logic)



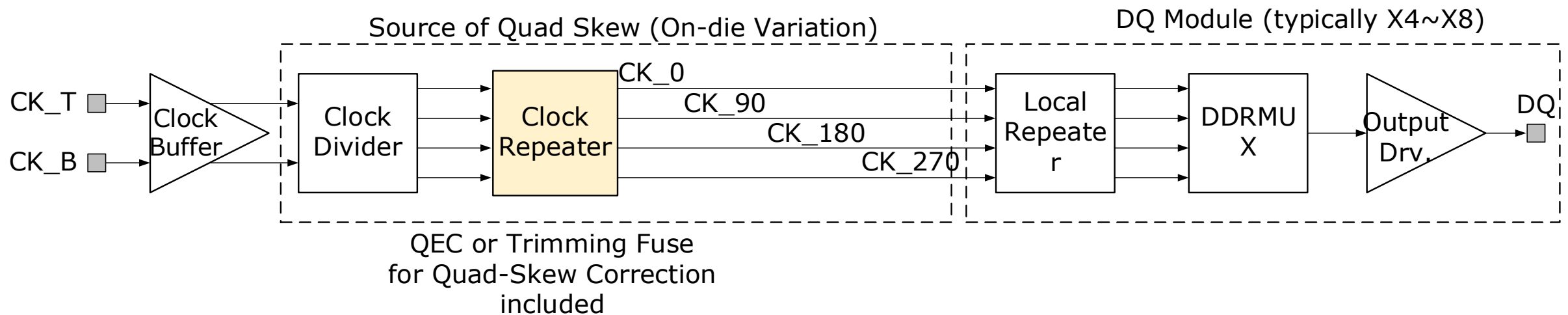
□ CMOS (Inverter-based)



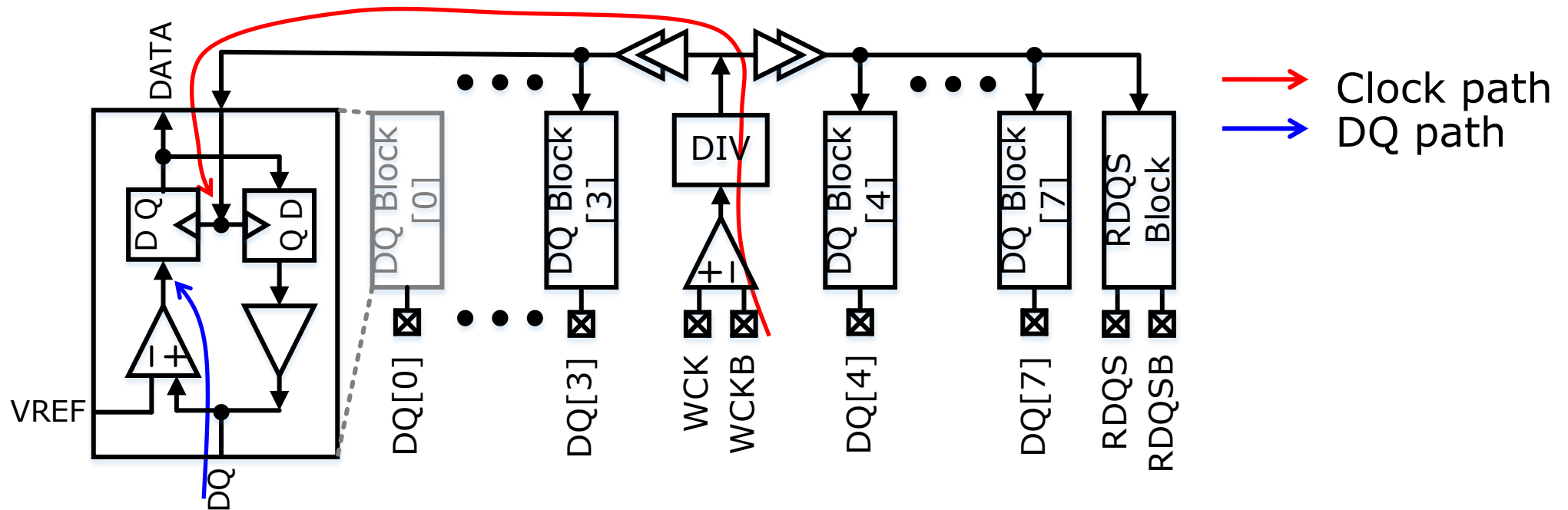
	CML	CMOS
Operating Frequency	High	Low
Power Supply Noise	Less sensitive	Highly sensitive (IVC can improve!!) *IVC : Internal Voltage Converter
Power Consumption	Large static	Only dynamic (transition)

Clocking (5) : 4-Phase

- 4-phase Clocking (a.k.a 2N clocking) :
 - Frequency divider is commonly used in DRAM internal clock tree in order to overcome the speed limit in transistors and routing metals.
 - However, 4-phase clocking is vulnerable to on-die variation. (Quad-Skew)
Recent DRAM's use QEC (Quad-Skew Corrector) or per-die calibration to minimize its impact.



RX Clocking (1)

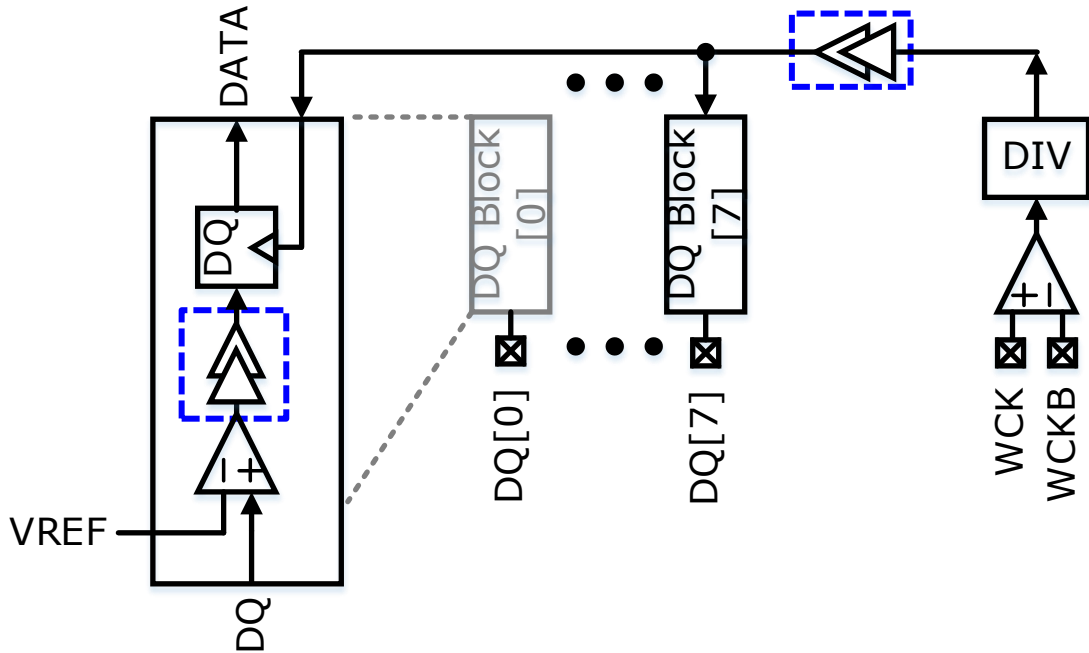


Example of Clock Tree and DQ in/out diagram

- WCK(CK) path is normally longer than DQ(CA) path due to the distribution delay to multiple DQs.
 - Need to consider the delay difference between clock and DQ.

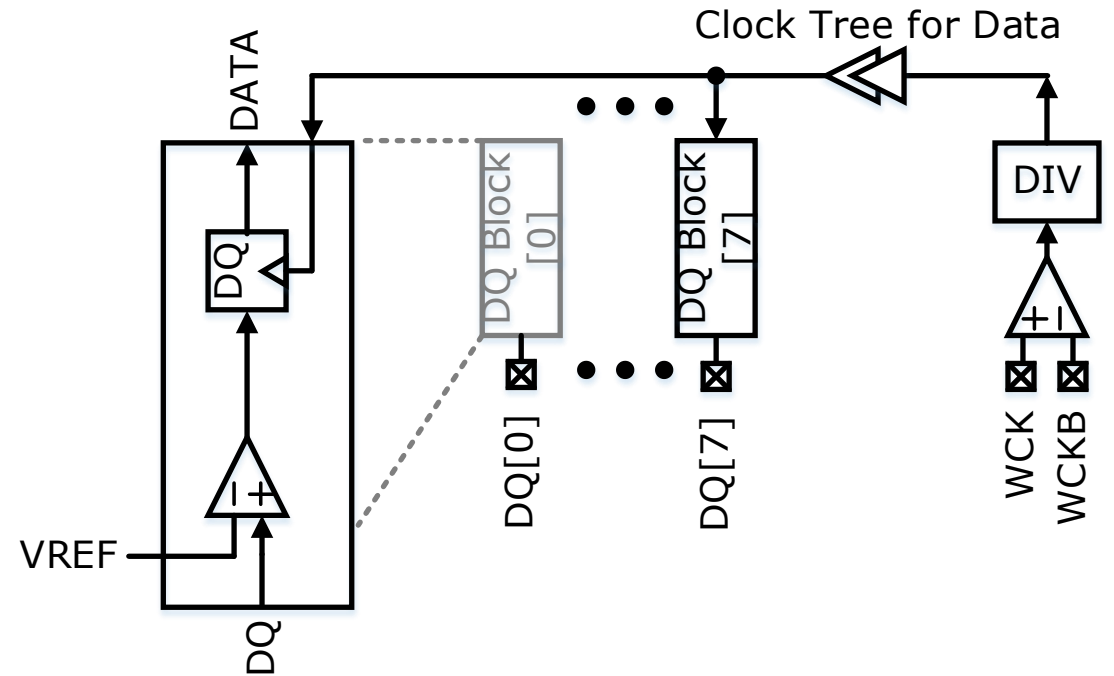
RX Clocking (2)

Matched Clocking



- DQ path delay to match clock path → Additional routing and replica delay.
- Simple timing but, performance degradation → LPDDR3, DDR4, ...

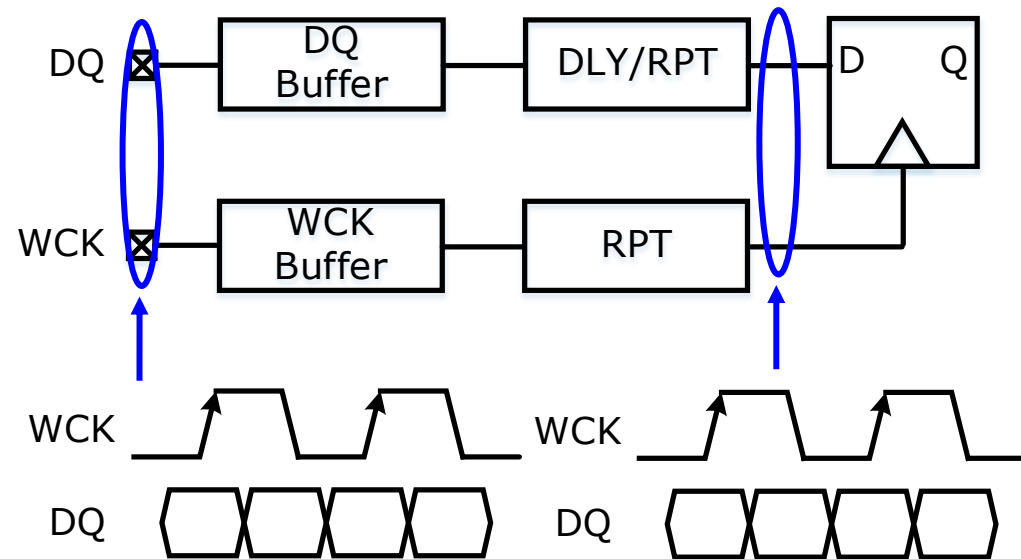
Un-Matched Clocking



- DQ path doesn't include delay path → No additional routing and replica delay.
- Complex timing but, good performance → GDDR5, LPDDR4, and DDR5

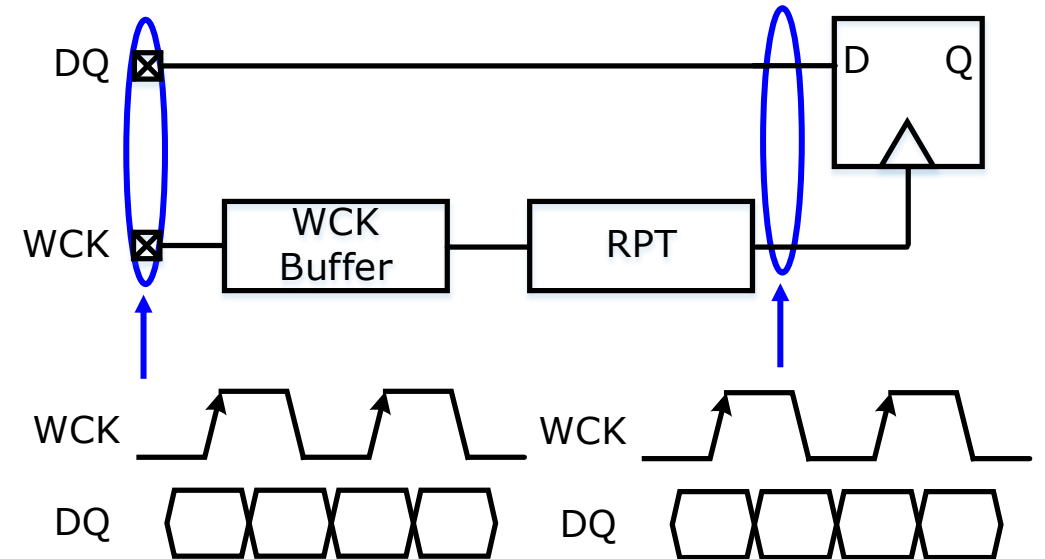
RX Clocking (3)

Matched Clocking



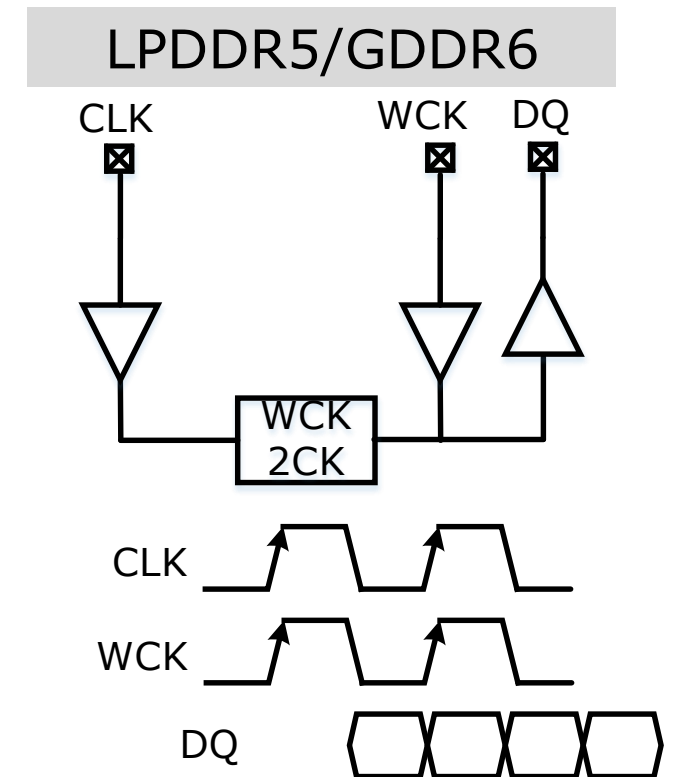
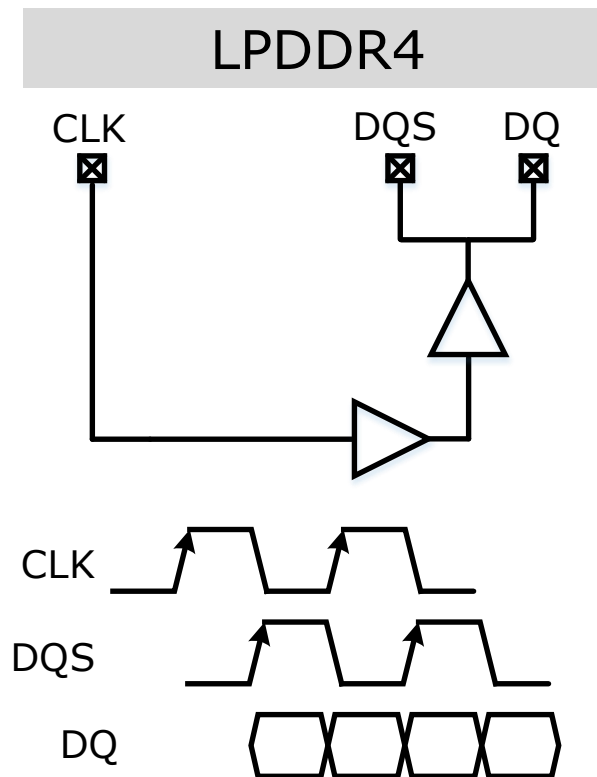
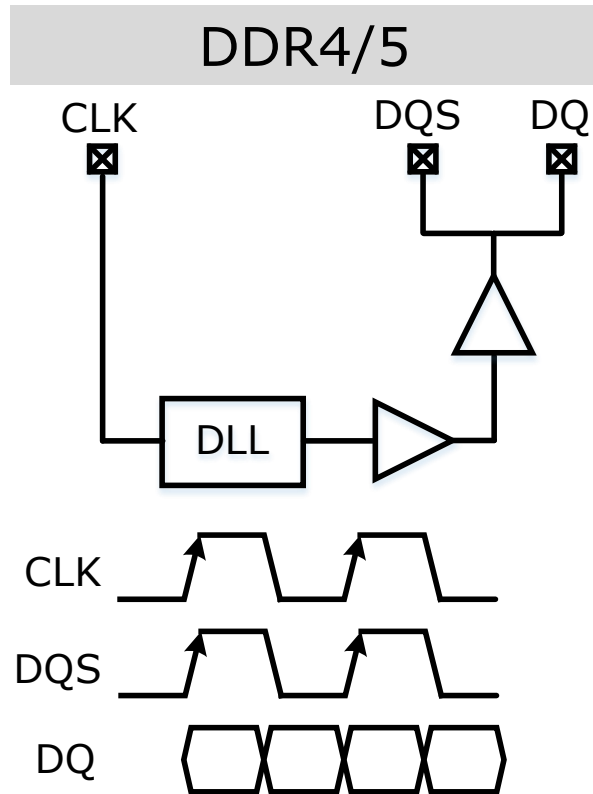
- Same timing relationship between DQ and WCK at DRAM PAD input and internal Receiver input

Un-Matched Clocking



- Different timing relationship between DQ and WCK at DRAM PAD input and internal Receiver input → Training!

TX Clocking



- Different timing relationship between CLK, DQS (WCK), and DQ for each applications
- DLL guarantees the aligned edge of DQS to CLK.

Pin Description and PKG Ball Map

3.4 Pad definition & Description

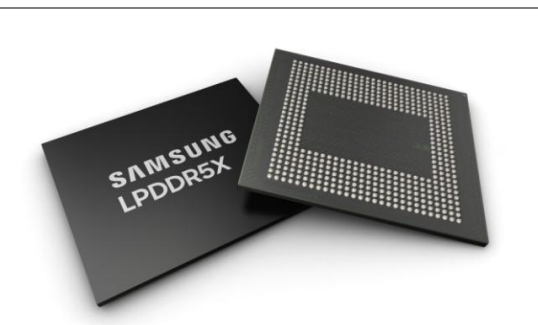
[Table 1] Pad Definition and Description

Symbol	Type	Description	NOTE
CK_t, CK_c	Input	Clock: CK_t and CK_c are differential clock inputs. All Double Data Rate (DDR) Command/Address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single Data Rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).	
CS	Input	Chip Select: CS is part of the command code, and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or Deep Sleep mode where it becomes an asynchronous signal.	
CA[6:0]	Input	Command/Address Inputs: CA signals provide the Command and Address input according to the Command Truth Table.	
DQ[15:0]	I/O	Data Input/Output: Bi-direction data bus.	
WCK[1:0]_t WCK[1:0]_c	Input	Data Clocks: WCK_t and WCK_c are differential clocks used for WRITE data capture and READ data output.	
RDQS [1:0]_t RDQS[1:0]_c	RDQS_t: I/O, RDQS_c: Output	Read Data Strobe: RDQS_t and RDQS_c are the differential output clock signals used to strobe data during a READ operation. And RDQS_t is also used as a Parity pin at Write with Link Protection enabled.	1
DMI[1:0]	I/O	Data Mask Inversion: DMI achieves multiple function such as Data Mask (DM), Data Bus Inversion (DBI), and Parity at read with ECC operation by setting the Mode Register and DMI is a bi-directional signal and each byte of data has a DMI signal.	1
ZQ	Reference	ZQ: ZQ is used to calibrate the output drive strength and the termination resistance as calibration reference. There is one ZQ pad per die. The ZQ pin shall be connected to VDDQ through a 240Ω ±1% resistor.	
VDDQ, VDD1, VDD2H, VDD2L	Supply	Power Supplies: Isolated on the die for improved noise immunity.	
VSS	GND	Ground Reference: Power supply ground reference	
RESET_n	Input	RESET: When asserted LOW, the RESET_n signal resets the die. Reset_n is an asynchronous signal.	

3.2 LPDDR5 SDRAM Package Ballout

496Ball FBGA																
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	NC	NC	VDD1	VDD2H	VDD3	VDDQ	VSS	VDD2H	VDD3	VDD2H	VDD3	VDDQ	VDD1	VDD3	VDD3	VDD1
B	NC	DQ3_A	RDQS0_T_A	VSS	DQ5_A	VSS	NC	VSS	VSS	DQ13_A	VSS	RDQS1_T_A	VSS	DQ1_A	VSS	RDQS0_C
C	DQ1_A	VSS	RDQS0_C_A	DM1_A	VDDQ	CA3_A	CS_A	CA4_A	CA5_A	VDDQ	DM1_A	RDQS1_C_A	DQ11_A	VDDQ	VDDQ	DQ3_C
D	VDDQ	DQ2_A	VDDQ	VSS	DQ6_A	VSS	CA2_A	CA3_A	VSS	DQ14_A	VSS	VDDQ	VSS	DQ2_A	DQ0_C	VSS
E	DQ8_A	VDDQ	WCK0_C_A	DQ4_A	VDDQ	CA1_A	VDD2H	VDD3	CA5_A	VDDQ	DQ12_A	WCK1_C_A	DQ10_A	VDDQ	VDDQ	DQ2_C
F	VSS	VSS	WCK0_T_A	VSS	DQ7_A	VSS	CK_T_A	CK_C_A	VSS	DQ15_A	VSS	WCK1_T_A	VSS	ZQ_A_C	RESET_n	VSS
G	VDD2H	VDD1	VDD3	VDDQ	VDD2H	VDD3	VSS	VSS	VDD2H	VDD3	VDD2H	VSS	VDD3	VDD2H	VDD3	VSS
H	VDD2H	VDD1	VDD2H	VDD2H												
J	VSS	VSS	VSS	VSS												
K	VSS	VSS	VSS	VSS												
L	VSS	VSS	VSS	VSS												
M	VSS	VSS	VSS	VSS												
N	VSS	VSS	VSS	VSS												
P	VSS	VSS	VSS	VSS												
R	VSS	VSS	VSS	VSS												
T	VSS	VSS	VSS	VSS												
U	VSS	VSS	VSS	VSS												
V	VSS	VSS	VSS	VSS												
W	VSS	VSS	VSS	VSS												
Y	VDD2H	VDD1	VDD2H	VDD2H												
AA	VDD2H	VDD1	VDD3	VDDQ	VDD2H	VDD3	VSS	VSS	VDD2H	VDD3	VSS	VDD3	VDD2H	VDD3	VSS	VSS
AB	VSS	VSS	WCK1_T_B	VSS	DQ16_B	VSS	CK_C_B	CK_T_B	VSS	DQ7_B	VSS	WCK0_T_B	VSS	RFU	ZQ_B_D	VSS
AC	DQ8_B	VDDQ	WCK1_C_B	DQ12_B	VDDQ	CA5_B	VDD3	VDD2H	CA1_B	VDDQ	DQ4_B	WCK0_C_B	DQ2_B	VDDQ	VDDQ	DQ10_D
AD	VDDQ	VDDQ	VSS	VSS	CA3_B	CA2_B	VSS	DQ9_B	VSS	VDDQ	VSS	DQ0_B	VSS	VDDQ	VSS	DQ14_D
AE	DQ8_B	VSS	RDQS1_C_B	DM1_B	VDDQ	CA5_B	CA4_B	CS_B	CA0_B	VDDQ	DM0_B	RDQS0_C_B	DQ3_B	VDDQ	VDDQ	DQ11_D
AF	NC	DQ11_B	RDQS1_T_B	VSS	DQ13_B	VSS	VSS	NC	VSS	DQ6_B	VSS	RDQS1_T_B	VSS	DQ1_B	DQ0_B	VSS
AG	NC	NC	VDD1	VDD2H	VDD3	VDDQ	VDD2H	VSS	VDD3	VDD2H	VDD2H	VDDQ	VDD1	VDDQ	VDD2H	VDD3

[Top View]



LPDDR5X Example

DRAM Specification

- JEDEC (Joint Electron Device Engineering Council) Solid State Technology Association
 - Semiconductor engineering standard of the American Electronics Industry Association (EIA) , which represents all sectors of the electronics industry.
- Specification
 - An explicit set of requirements to be satisfied by a product.

JEDEC STANDARD

**Low Power Double Data Rate
(LPDDR) 5/5X**

JESD209-5C
(Revision of JESD209-5B, June 2021)

June 2023

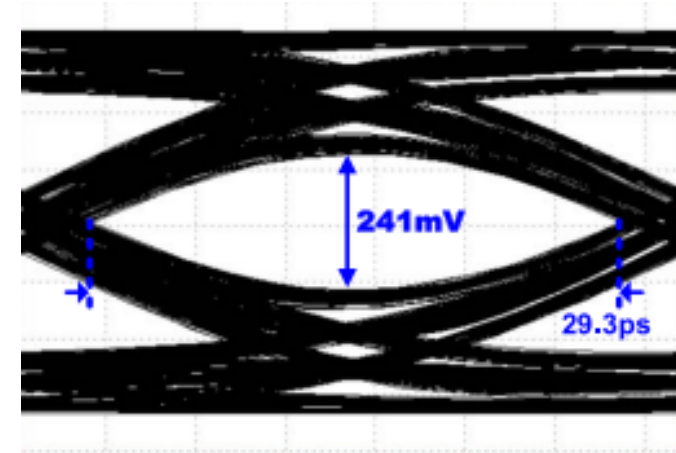
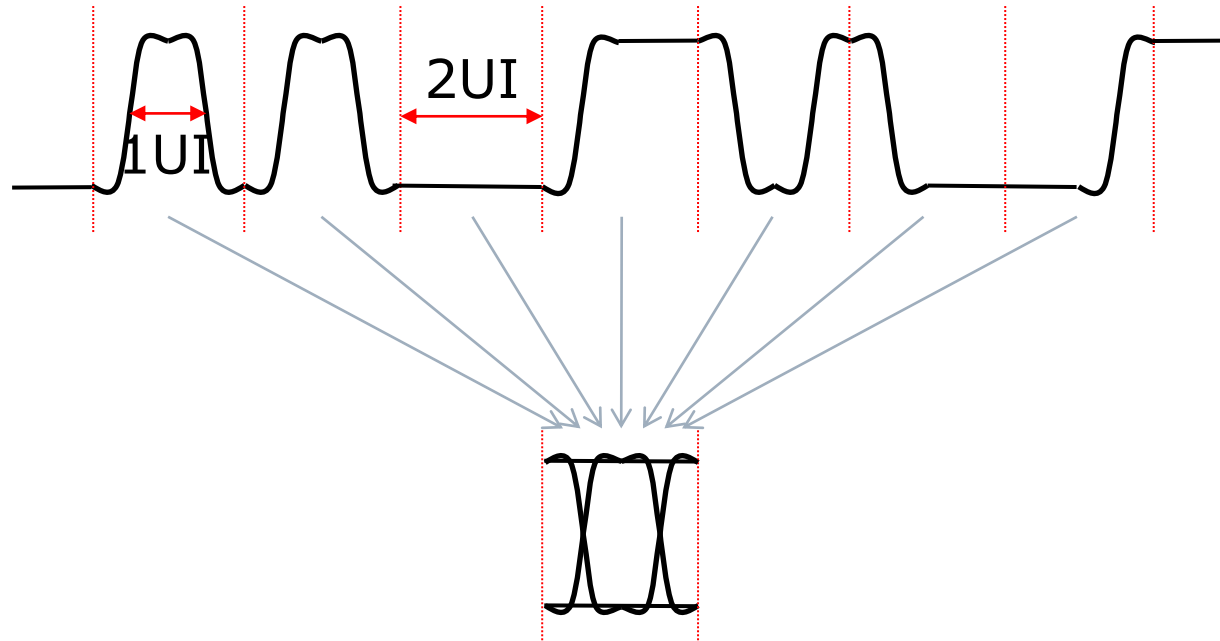
JEDEC SOLID STATE TECHNOLOGY ASSOCIATION



Outline

- DRAM I/O and Applications
 - DRAM Application
 - Signaling and Clocking
- **I/O Speed Extension**
 - **Equalization**
 - Training and Calibration
 - Link Protection
- Present and Future DRAM
- Conclusion

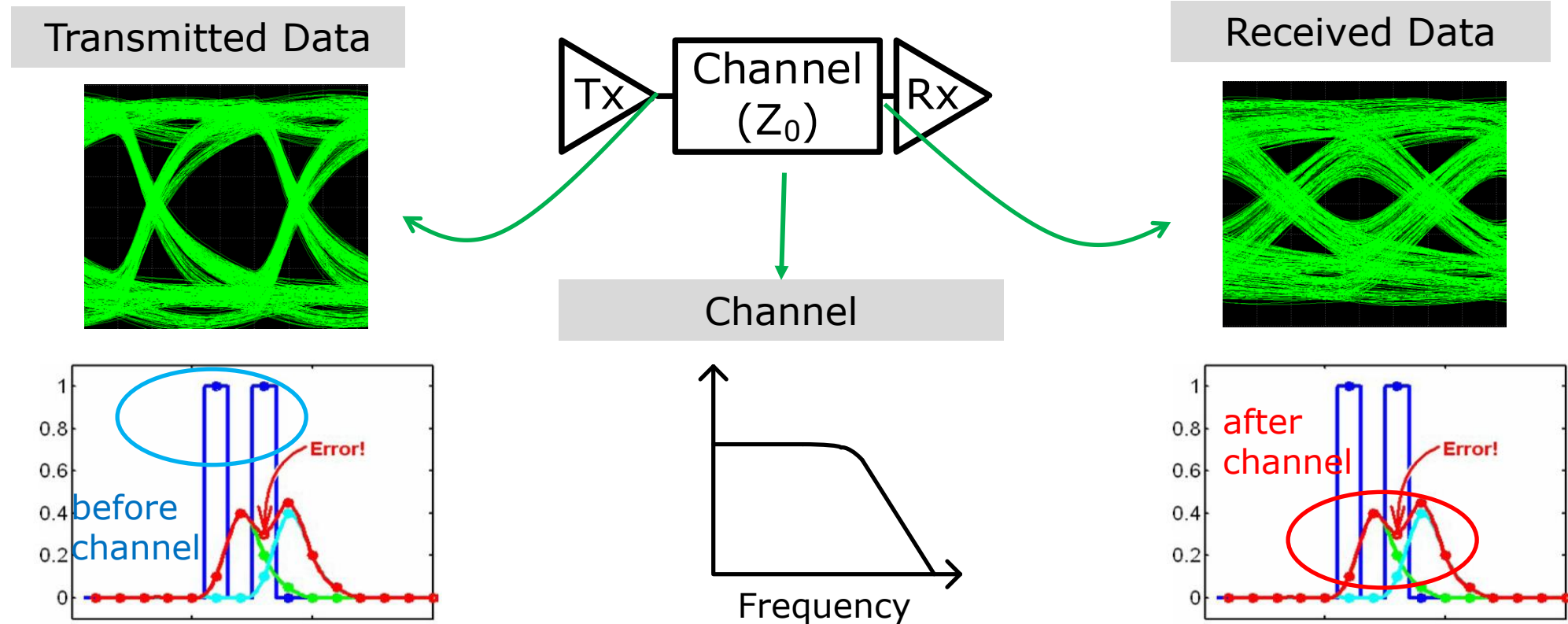
Eye Diagram?



D. Lee et al, doi: 10.1109/JSSC.2022.3222203.

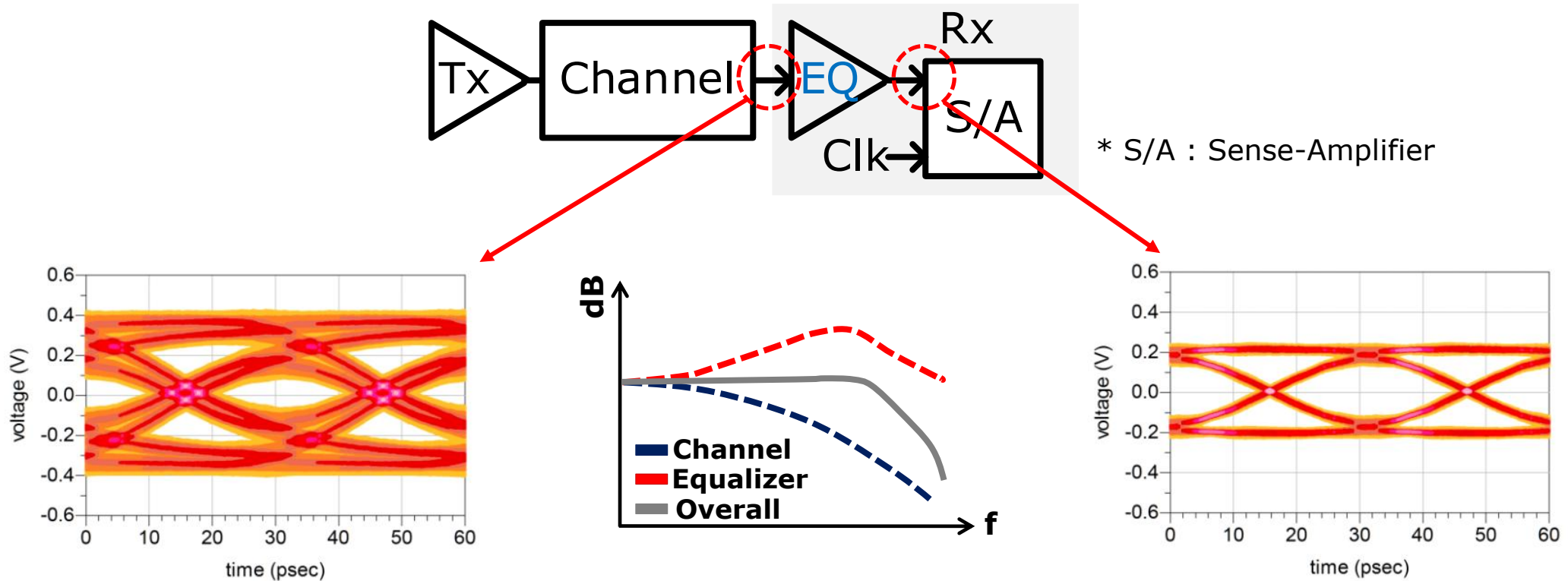
- ❑ Map all UI intervals on top of one and other
- ❑ Represent what a receiver will see & specify characteristics of a transmitter.

Why Equalization?



- Channel has low-pass response (Low Pass Filter-LPF-Characteristic) & Inter-Symbol Interference.
 - Difficult to restore the received data at high frequency.

Equalization



- Channel frequency dependent loss compensation.
→ High frequency boosting by equalizer

Channel and Application

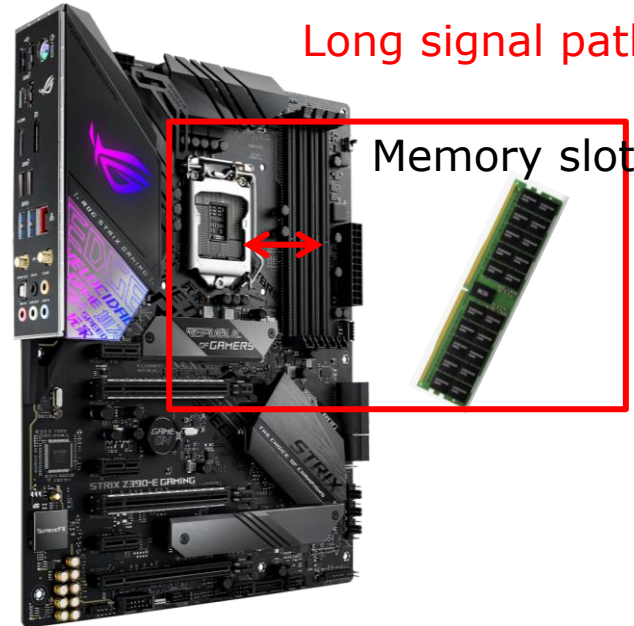
LPDDR

Small form factor



<https://www.samsung.com/sec/>

DDR



Long signal path

Memory slot

<https://www.intel.com>
<https://semiconductor.samsung.com/dram>

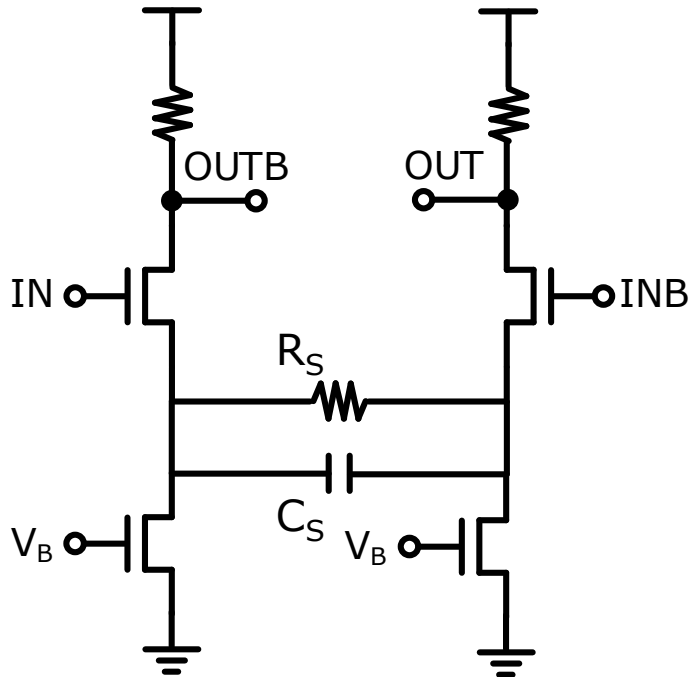
Channel loss

	Insertion loss (dB)
LPDDR	0.5 ~ 2
GDDR	2 ~ 3
DDR	5 ~ 10

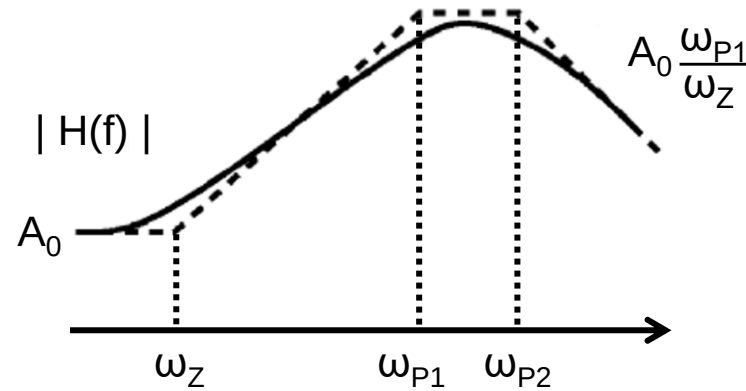
- ❑ Channel generally means signal path b/w DRAM and SOC (AP, GPU).
- ❑ Long signal path length = more loss.

CTLE (RX)

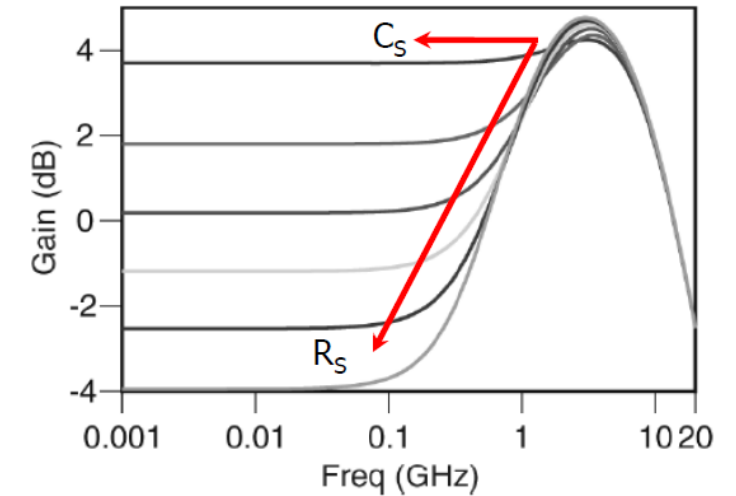
CTLE Circuit Diagram



CTLE Transfer Function, $H(f)$



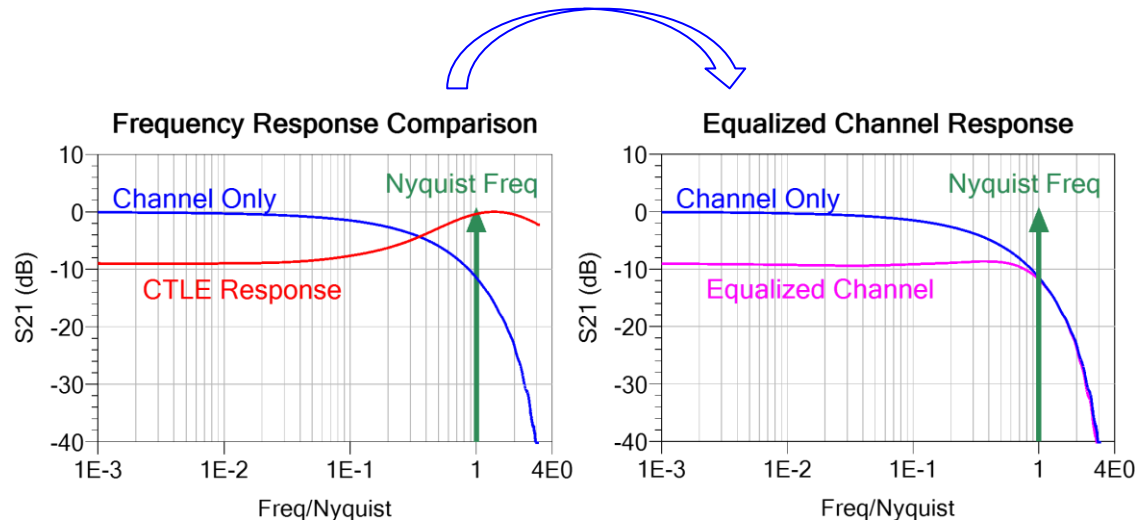
RS & CS vs. $H(f)$



- ❑ Continuous-Time Linear Equalizer
- ❑ Induce the high-frequency peaking to compensate the channel loss.
- ❑ Filter operation possible without clock synchronization

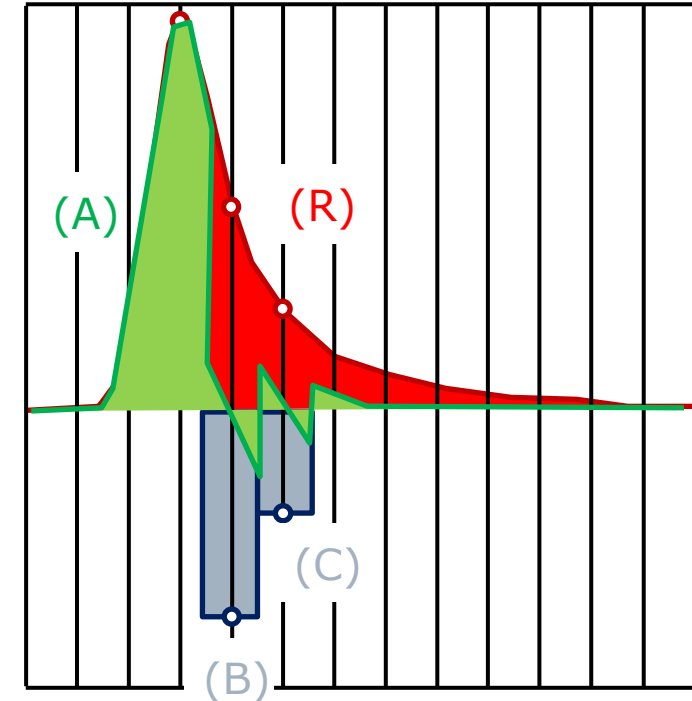
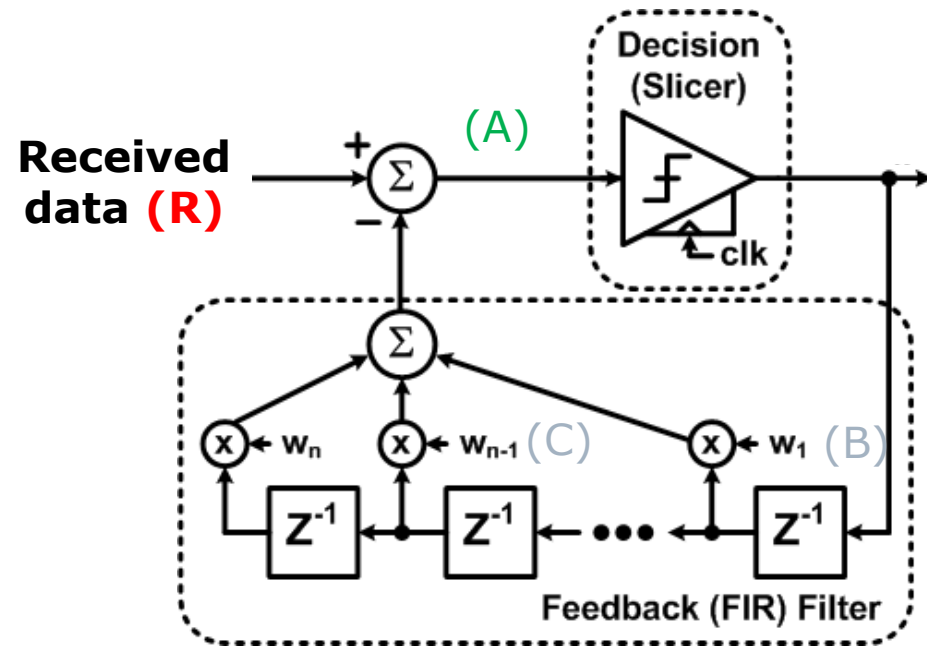
CTLE (RX)

Pros	Cons
✓ Capacitor is not very large ✓ Input matching is easily done by 50-ohm resistor	✓ DC gain of amp is degraded by resistive degeneration ✓ Amplifies noise ✓ PVT sensitivity



<https://docs.keysight.com/eesofapps/post-6-eye-opening-experience-with-ctle-678068358.html>

DFE (RX)

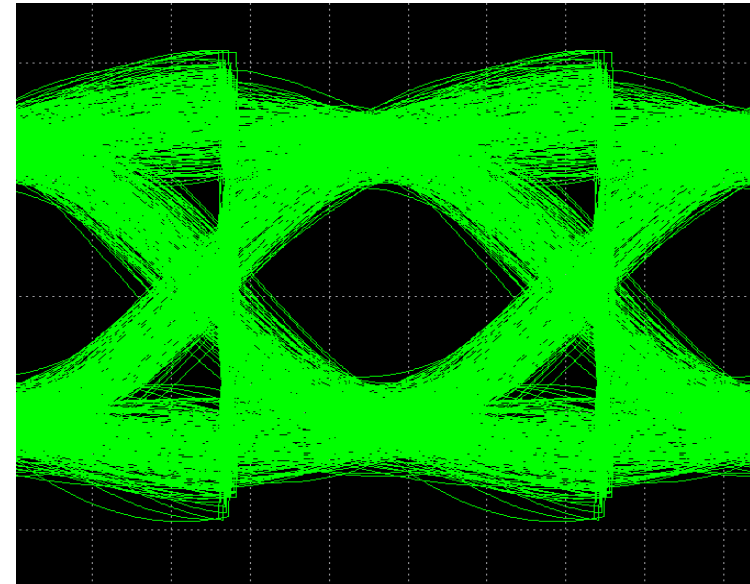
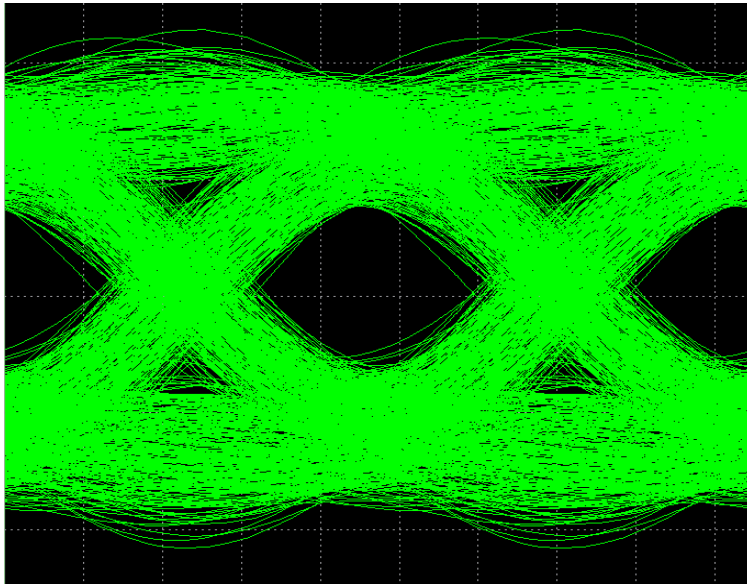


- ❑ Decision-Feedback Equalizer (Non-linear characteristic due to feedback)
- ❑ Slicer makes a symbol decision.
- ❑ ISI is then directly subtracted from the incoming signal via a feedback FIR filter.

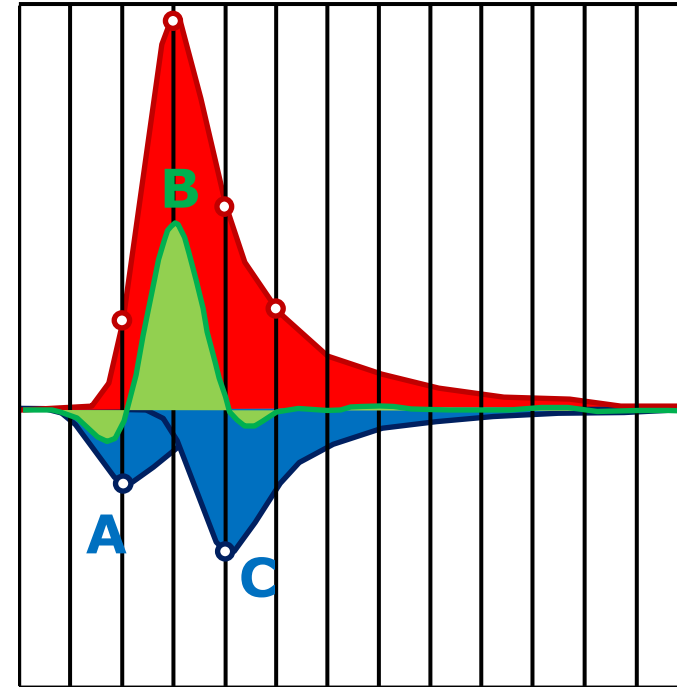
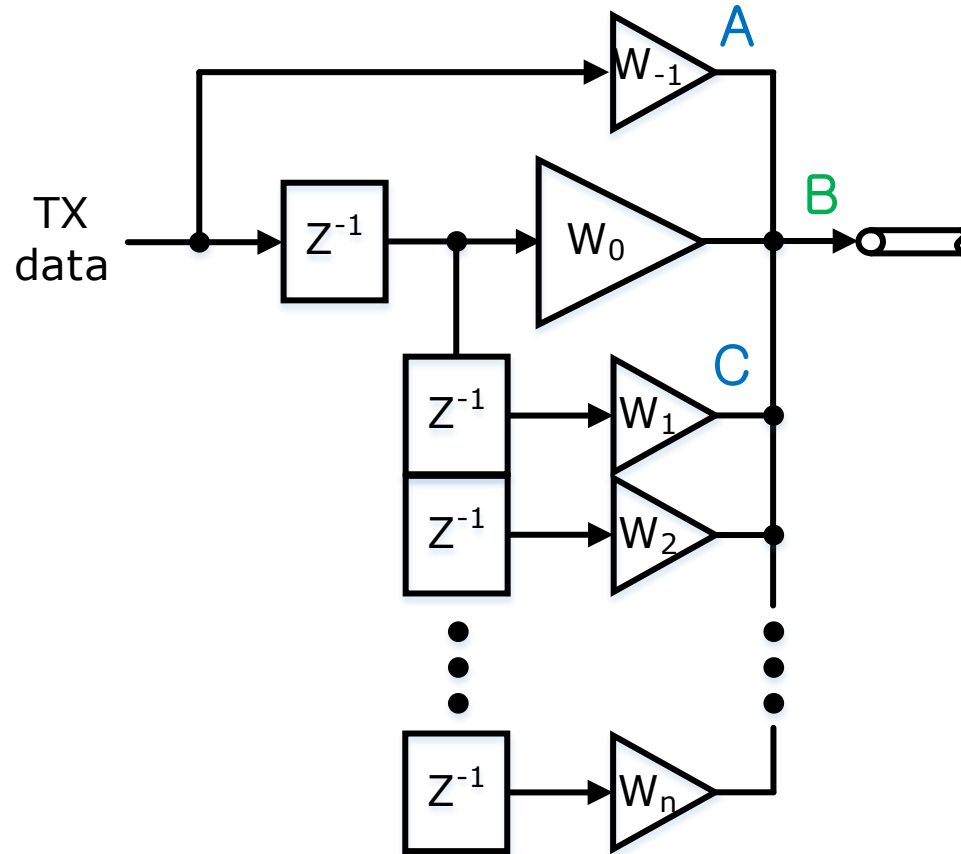
DFE (RX)

Pros	Cons
✓ No noise boosting	✓ Clock recovery necessary ✓ Difficult to open closed eye ✓ Pre-cursor cannot be removed ✓ Feedback-time requirement

** Assuming no Error Propagation



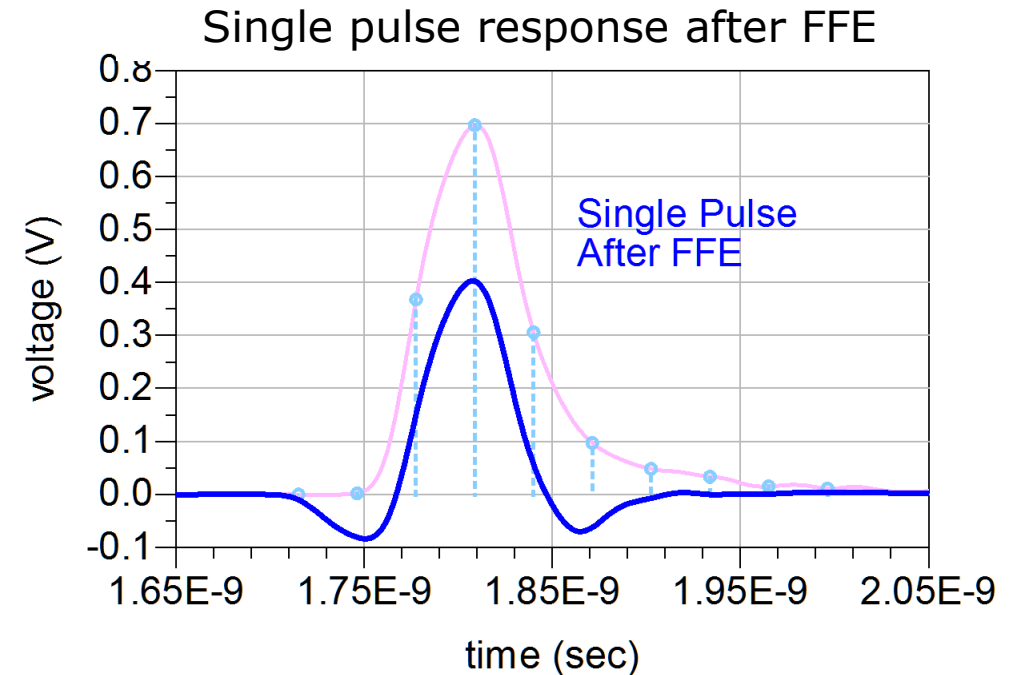
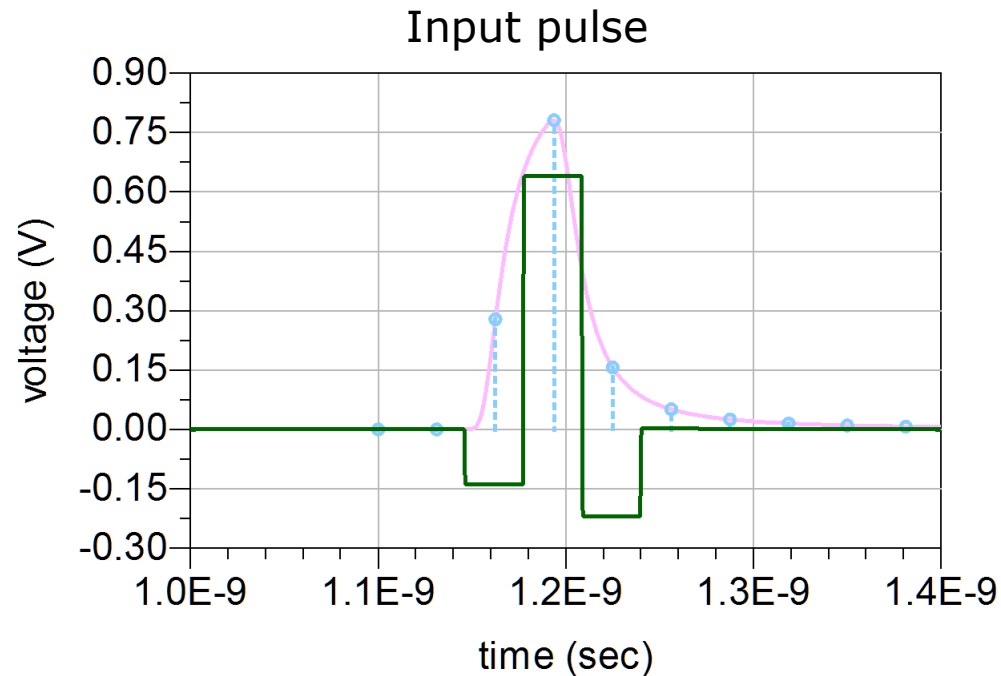
TX FFE



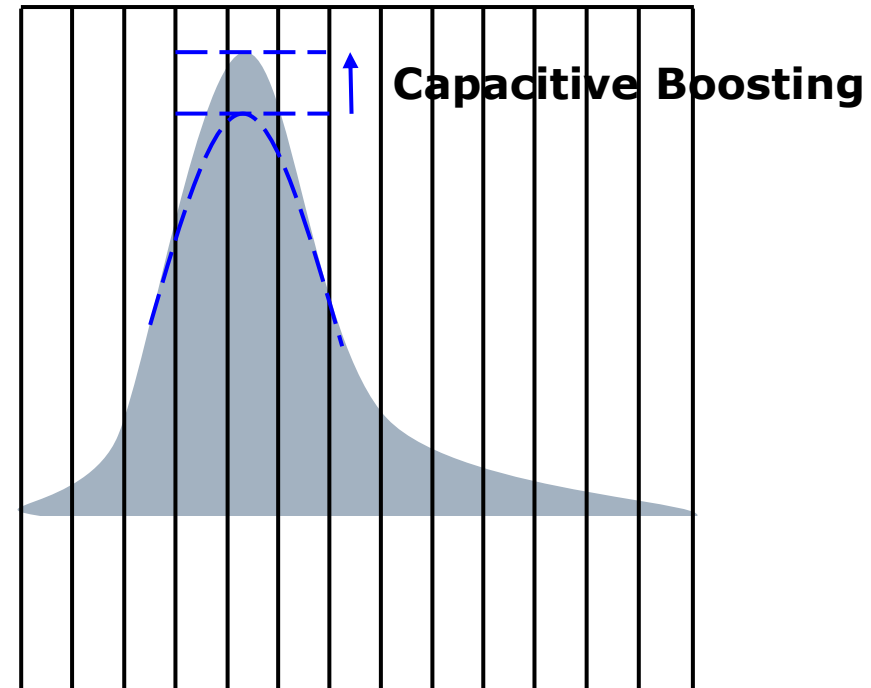
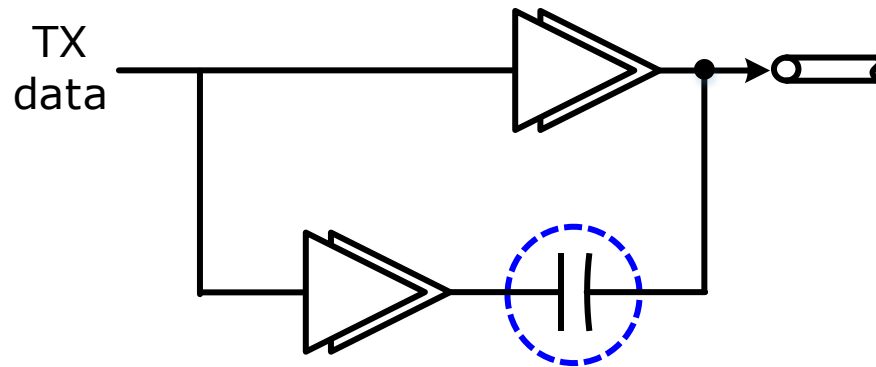
- ❑ Forcing cursors to 0 can be implemented by FIR filtering
- ❑ ISI can be removed since we know input data in TX-side

TX FFE

Pros	Cons
✓ Simple to implement ✓ Can cancel ISI in pre-cursor ✓ Don't amplify noise (digital nature)	✓ Attenuates low frequency content ✓ "feedback-channel" to tune filter taps



Capacitive EQ (TX)



- ❑ Pre-emphasis driver with boosting capacitor (MOS cap, MOM cap).
- ❑ Capacitive pre-emphasis can boost main cursor.

Equalization Comparison

- ❑ RX Equalization schemes are mandatory for the current-generation DRAMs in the specification regarding the speed and channel loss.

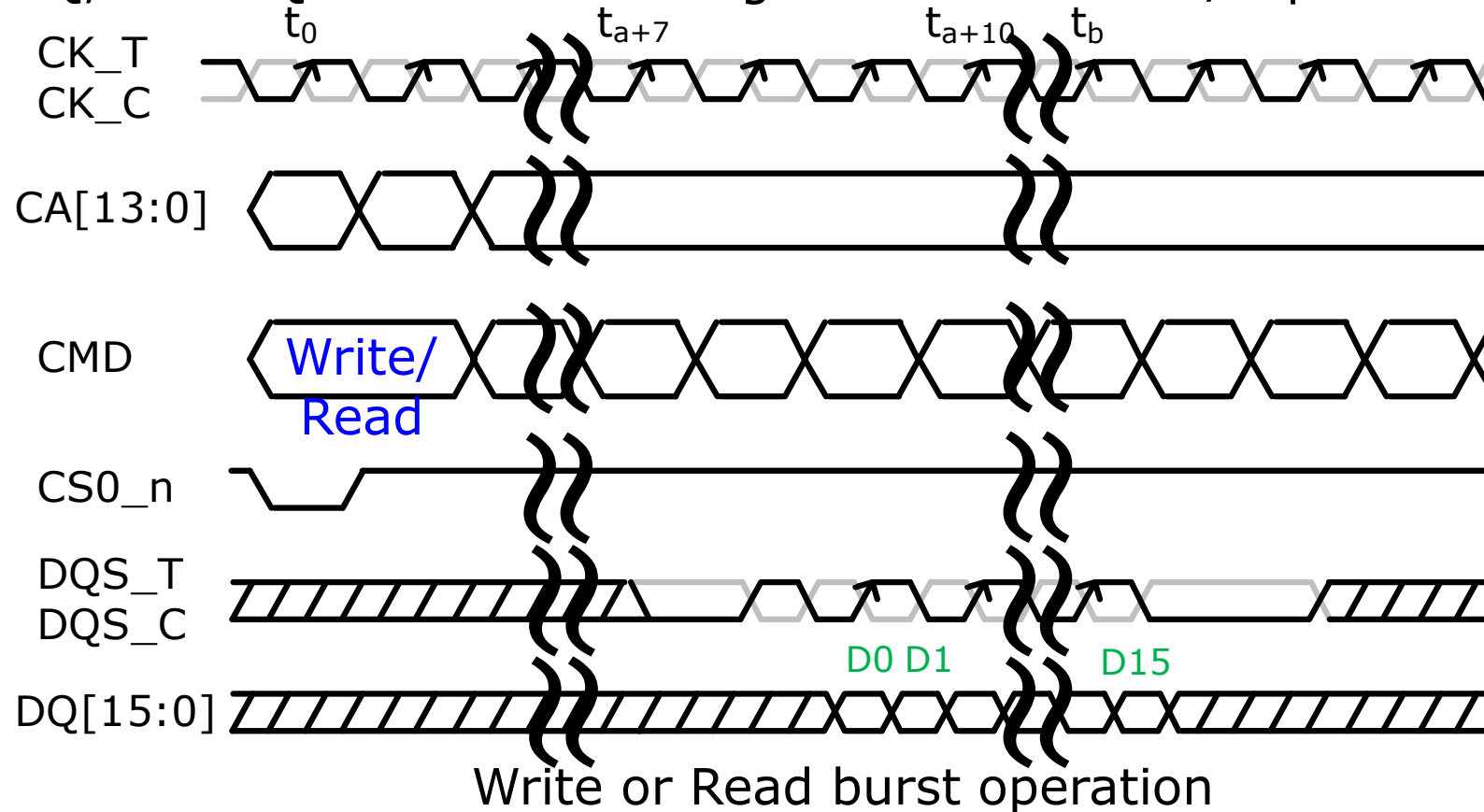
	D4	D5	LP4	LP5	G6	G7
Datarate [Gb/s]	3.2	6.4	3.2	6.4	16	32
Channel	Multi drop	Multi drop	Point to point	Point to point	Point to point	Point to point
Read	Vendor Specific	Vendor Specific	Vendor Specific	Vendor Specific	Optional	Require Controllability ** Type : Vendor-specific
Write	Vendor Specific	CTLE (Optional) + DFE	Vendor Specific	DFE	DFE	CTLE + DFE

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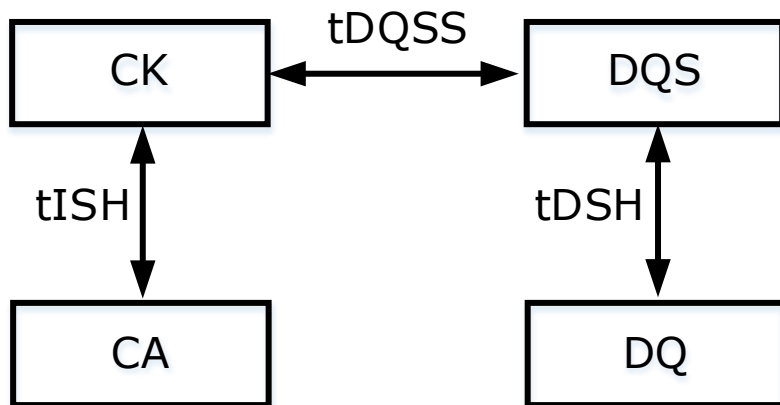
I/O during DRAM Operation

- DRAM write and read operations are defined based on the ideal timing for CK, CA, DQ, and DQS. → Is this timing same on DRAM I/O pins?



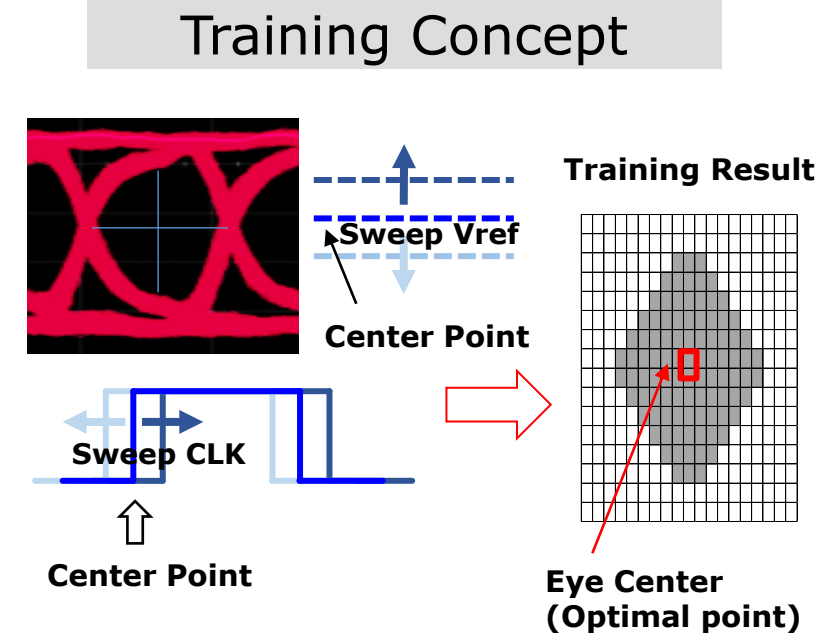
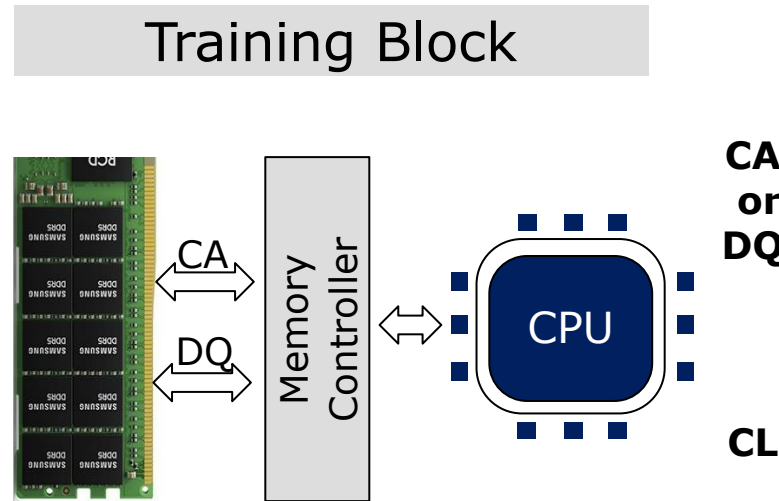
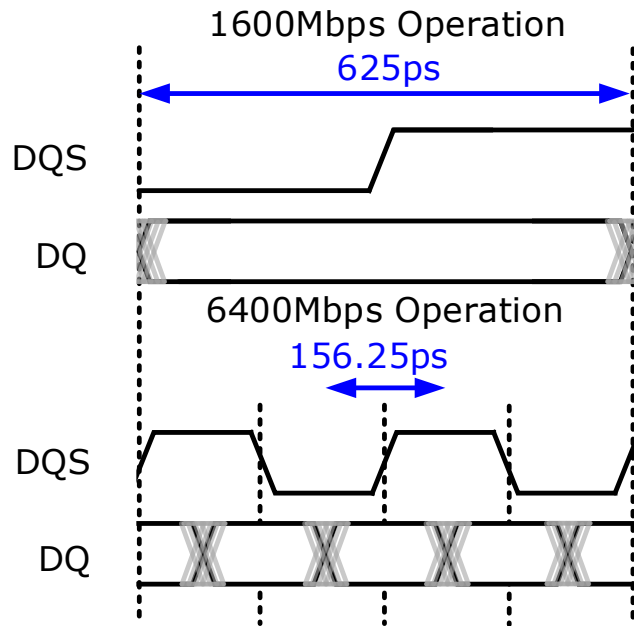
DRAM I/O AC Parameters

- ❑ The delay from GPU to DRAM cannot be matched for all DRAM I/O pins (CK, CA, DQ, and DQS), unlike the ideal timing diagram.
- ❑ The timing relationship between DRAM I/O pins are initially optimized through training operations (Mandatory for high-speed DRAMs).
- ❑ The timing constraints to guarantee the reliable operation of DRAM are defined in the specification.



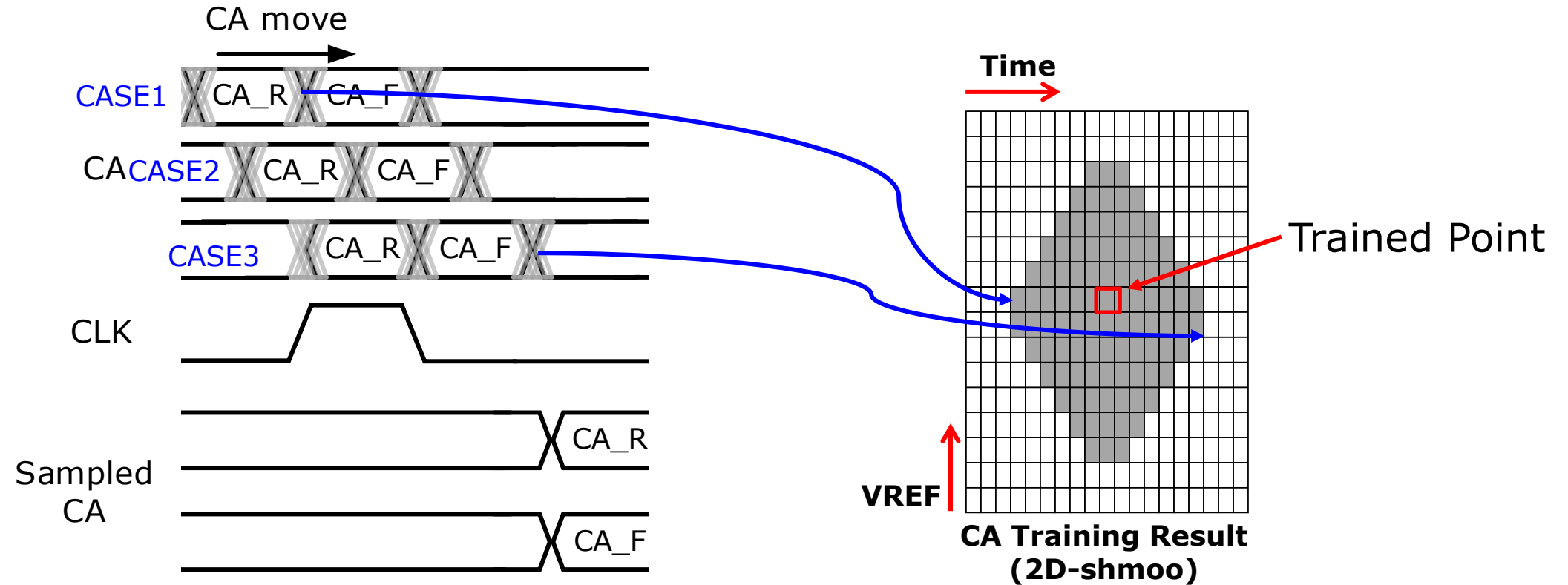
- ❖ SoC is required to maintain :
 - t_{DQSS} : CK-DQS $\rightarrow$ Write leveling
 - t_{DSH} : DQS-DQ (Write) $\rightarrow$ Write training
 - t_{ISH} : CK-CA $\rightarrow$ CA calibration (Command Bus Training)
 - t_{DQSQ} : DQS-DQ (Read) $\rightarrow$ Read training

Why Training?



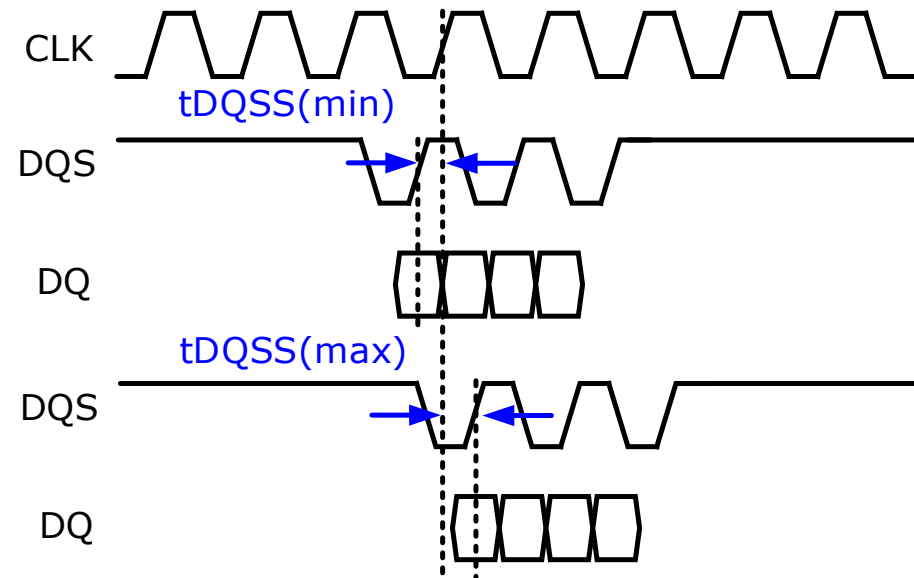
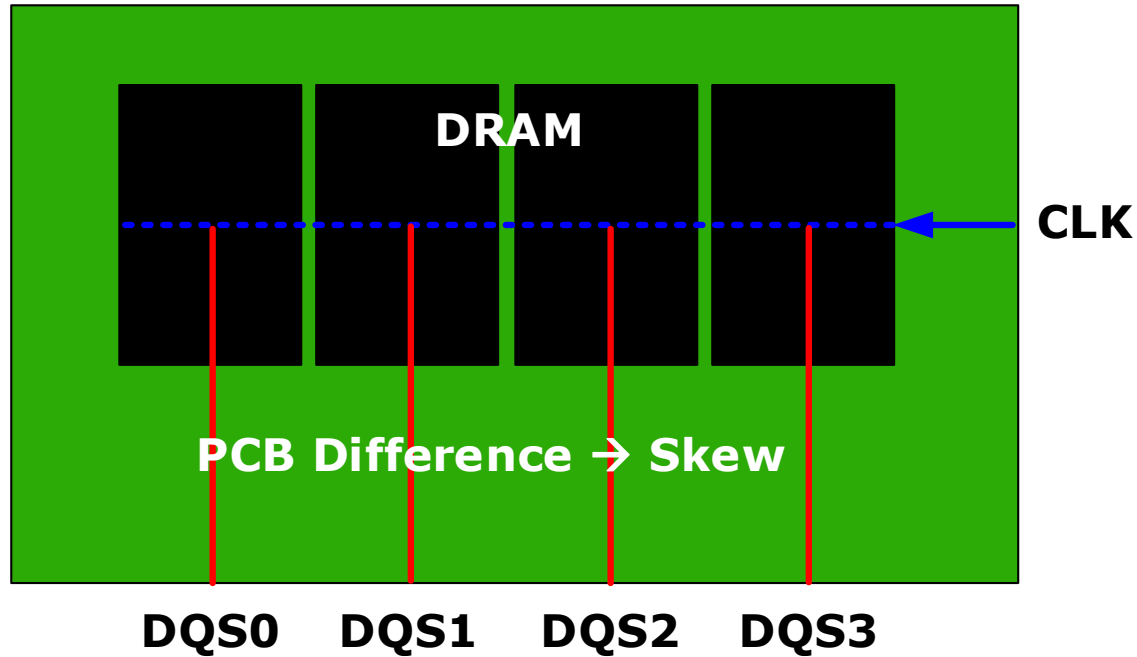
- ❑ Speed extension → Valid window smaller → Internal margins smaller.
- ❑ PCB and Board design restriction → Skew between pins not avoidable.
→ SoC (Memory controller or CPU) is required to search the optimal timing relationship for DRAM.

CA Training



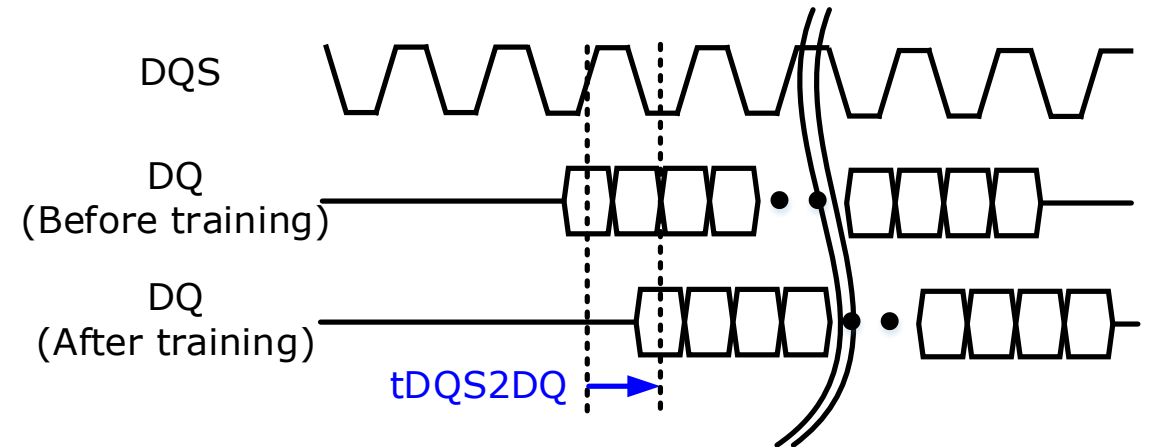
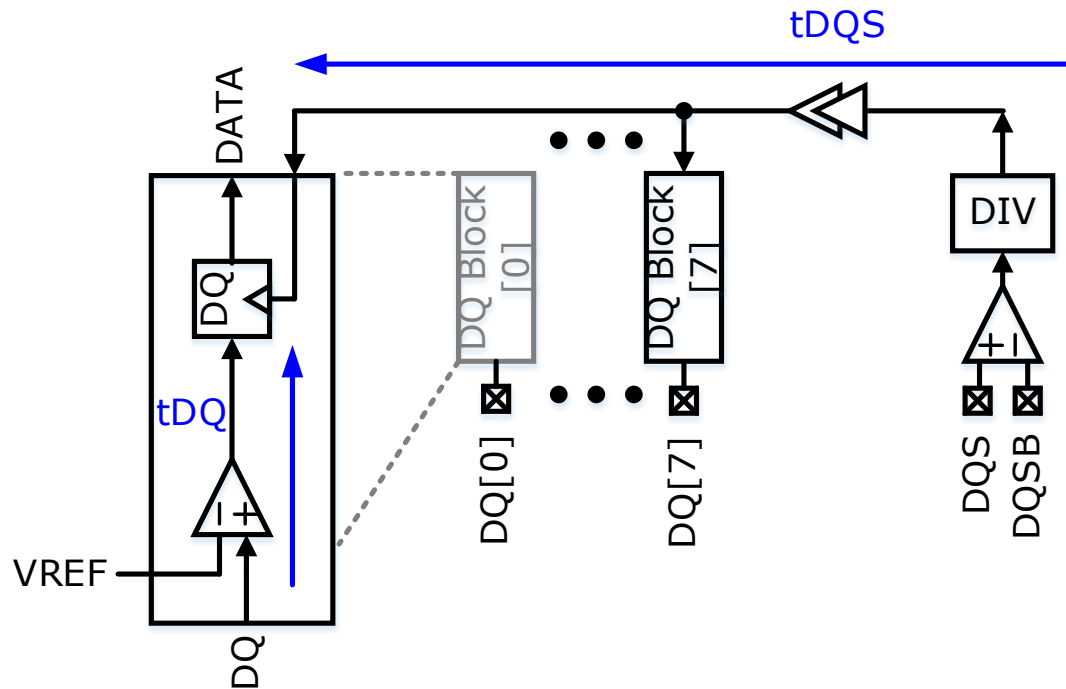
- ❑ SoC is supposed to adjust CA delay referring to CLK.
- ❑ Select the optimal point to maximize the timing margin during operations.

Write Leveling



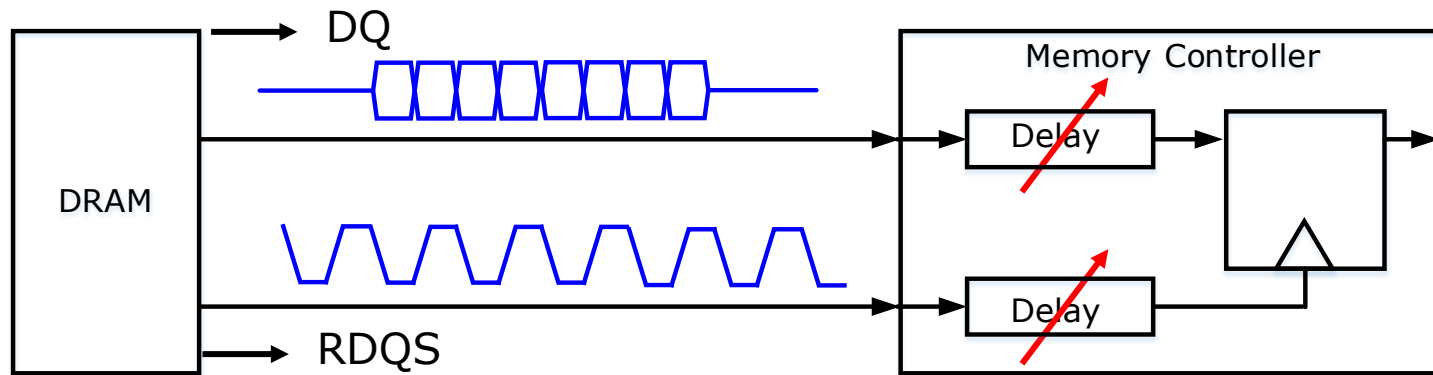
- ❑ Compensate the skew between CLK and DQS.
- ❑ Maximize tDQSS margin.

Write Training

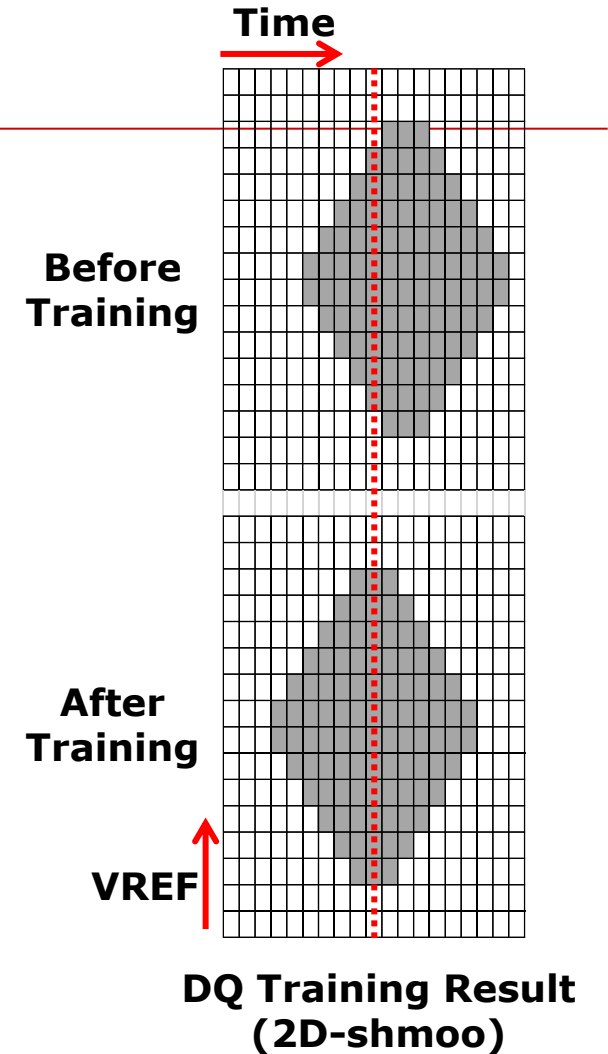


- ❑ Match the delay of DQ to that of DQS at receiver input → Add the delay to DQ timing in order to compensate DQS delay (t_{DQS}) inside DRAM.
- ❑ Maximize t_{DQSDQ} margin

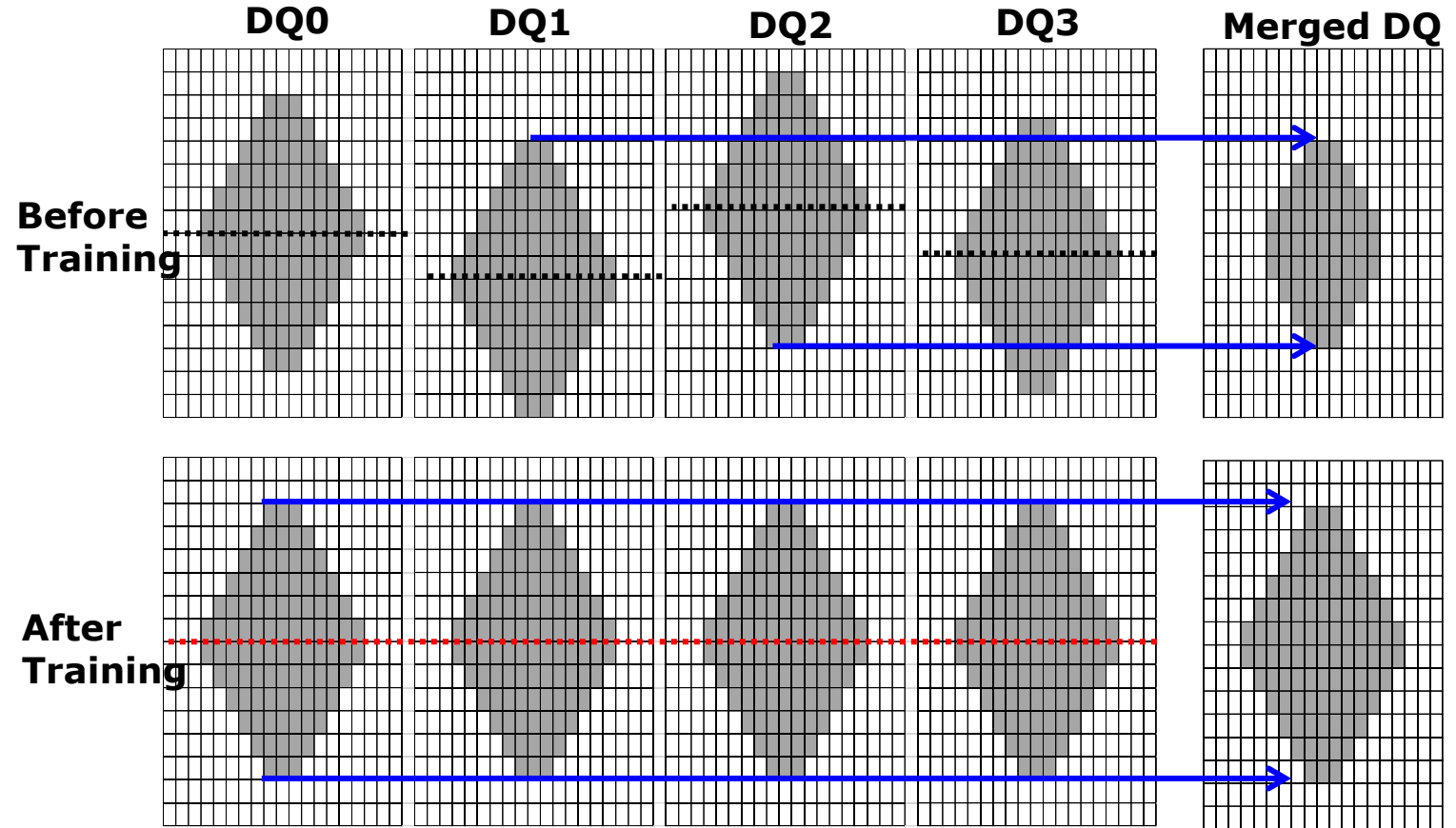
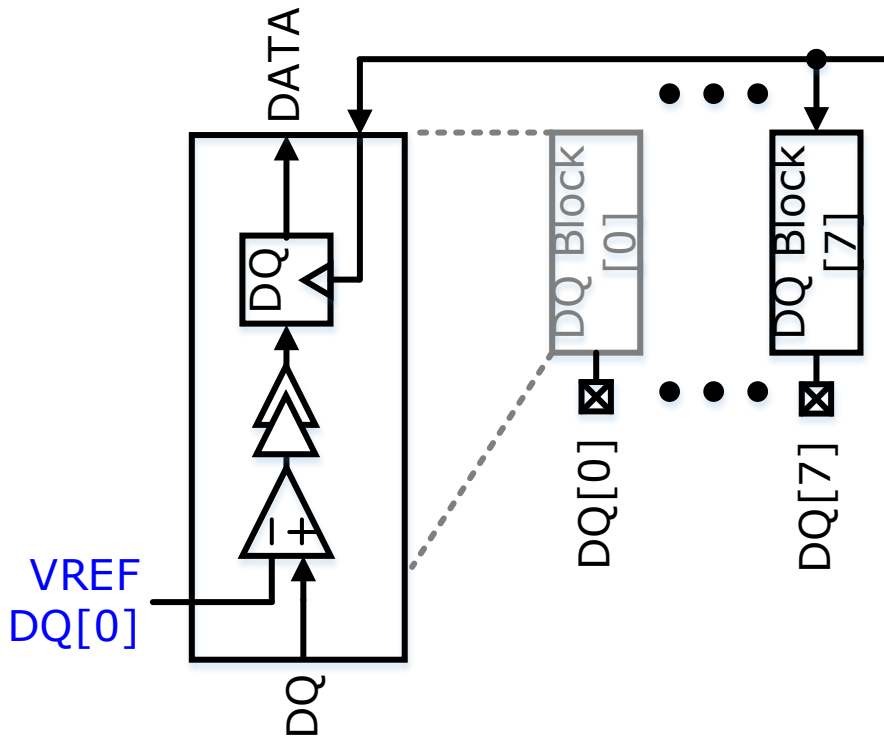
Read Training



- ❑ The similar operation to write training at SoC side.
- ❑ Compensate the channel skew among DQs.
- ❑ Allows SoC to search the centering of read data eye by optimizing SoC internal delay.



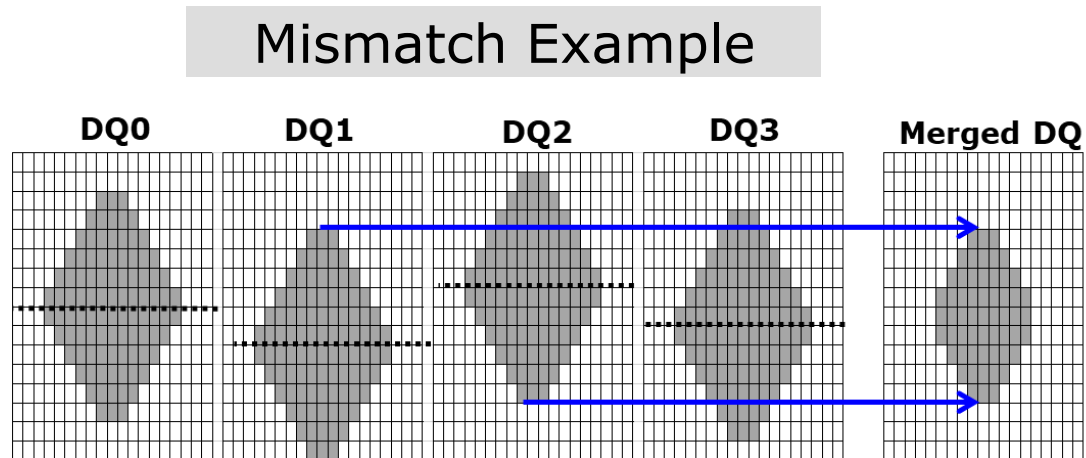
Advanced Training



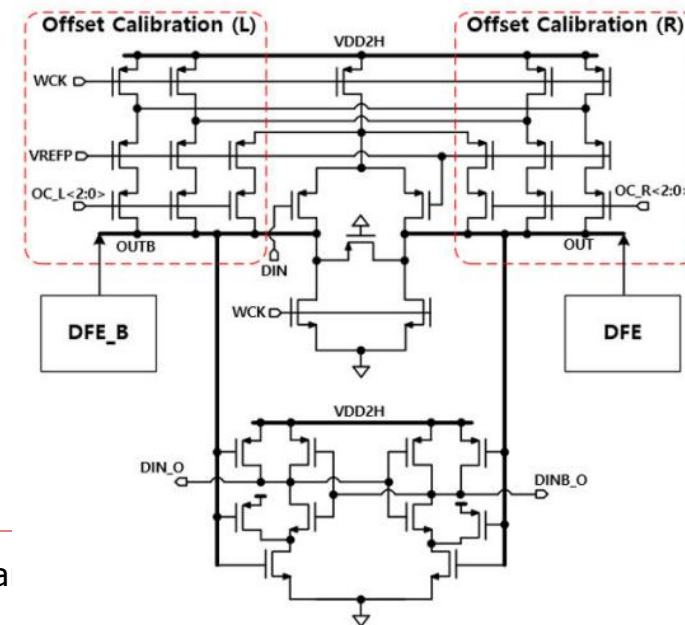
- Allows SoC to search the centering of per WR data eye by optimizing DRAM internal VREFD level.

Offset Cancellation

- Vref is shared between multiple DQ's to reduce stand-by power and training time in DRAM (Except GDDR6/7).
- The different mismatch in each sampler generated by the process and layout variation, causes write IO margin reduction (in the left figure).
- Offset calibration feature makes the input eye equal between DQ's.
 - Usually implemented by controlling input-transistor strength (in the right figure).

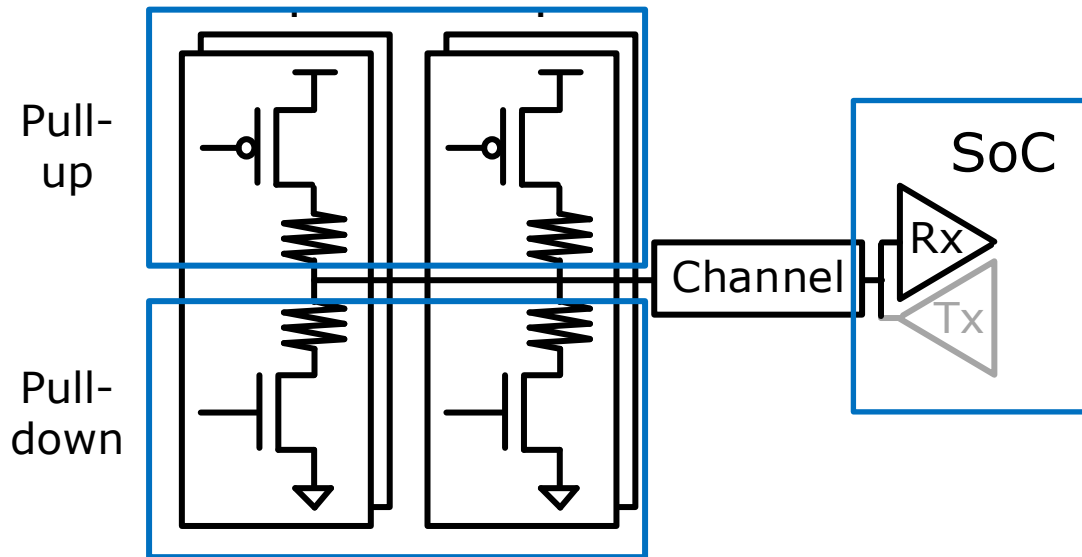


* AHN et al.: 1.01-V 8.5-Gb/s/pin 16-Gb LPDDR5x SDRAM WITH ADVANCED I/O CIRCUIT

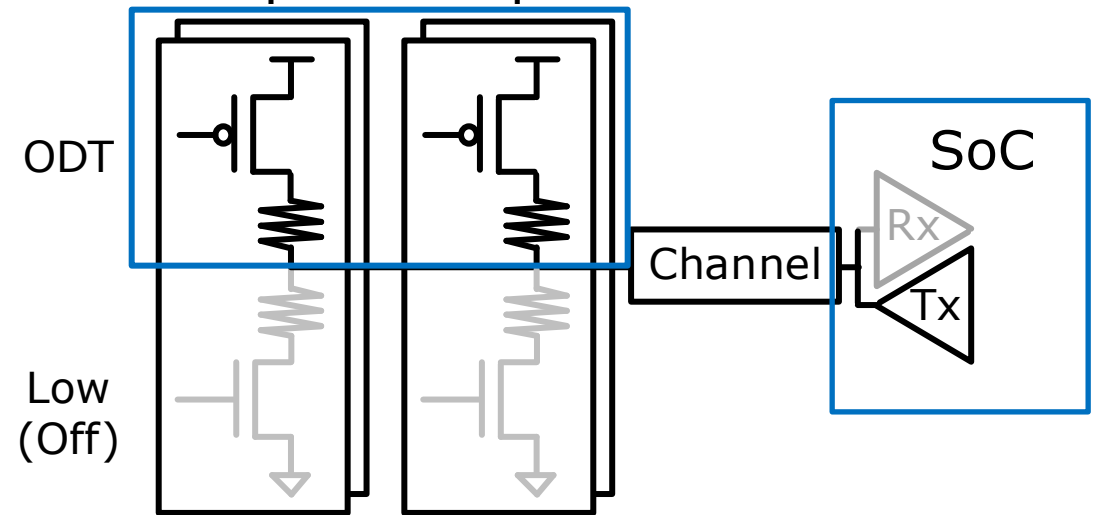


Output Driver & On-Die Termination

Output Driver



On-Die Termination

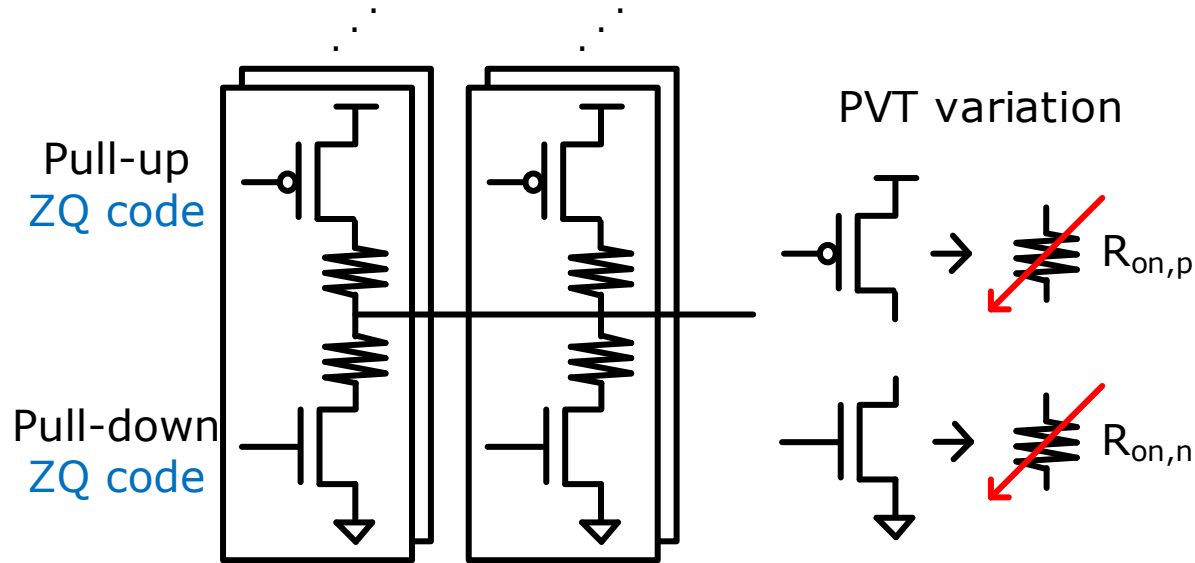


- Output Driver = Pull-up and Pull-down resistance during read operation
- On-die termination during write operation → Output pull-up driver used

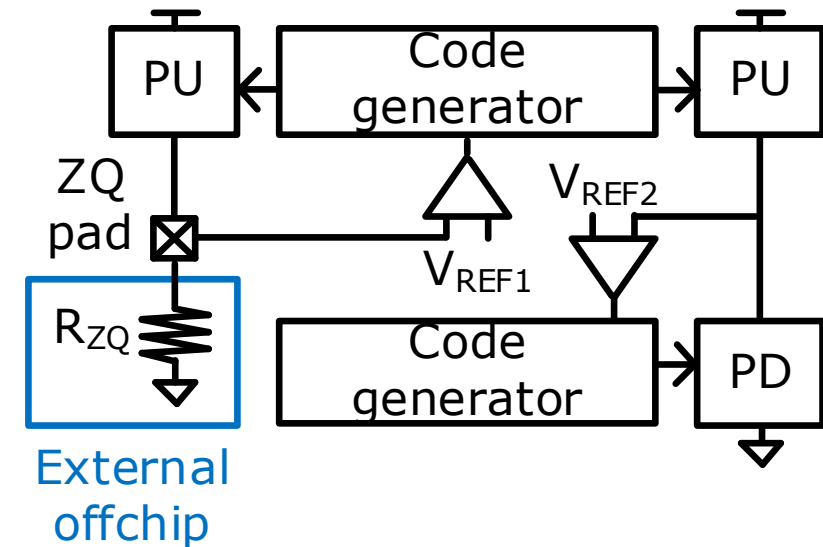
Why ZQ Calibration?

Impedance **ZQ** Quality

Output Driver of DQ circuit



Frontend of ZQ & External R_{ZQ}



- ZQ calibration guarantees constant R_{on} (Drivability) even with PVT (Process, Voltage, Temperature) variation by using external off-chip ZQ resistor which has the same resistance across PVT.

DCM/DCA

❑ DCM / DCA? Duty Cycle Monitor / Adjuster

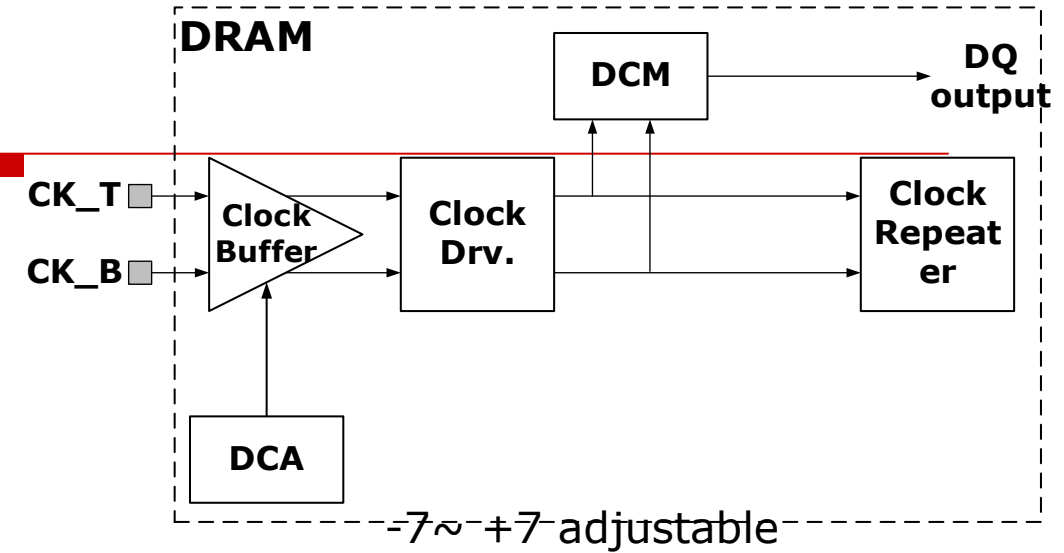
❑ Source of clock duty distortion :

- 1) Memory Controller PLL/Clock Driver
- 2) Not perfectly matched board/Pkg. PCB
- 3) DRAM Clock Buffer

❑ DRAM controller is able to correct clock duty by using DCM/DCA

→ Controller programs DCA mode registers in DRAM to adjust duty cycle.

It stops adjusting duty cycle, once DCM result switches.



Example) Programming Sequence → Stop

DCA Code	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7
DCM Output	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1

Switch

Outline

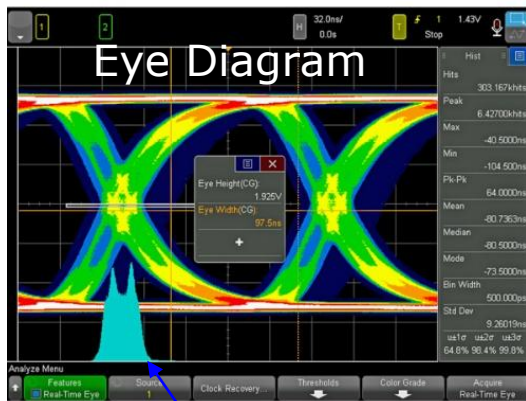
- DRAM I/O and Applications
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Jitter and BER Consideration

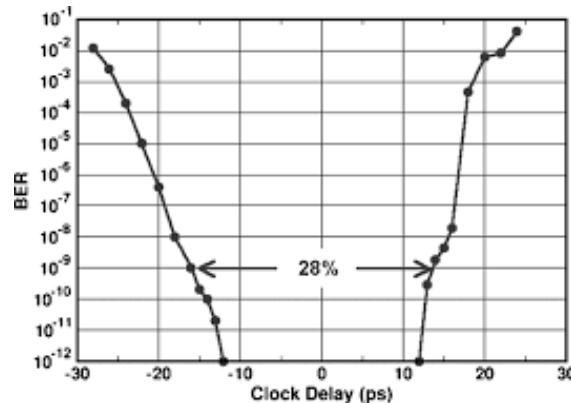
□ What is Jitter ?

- Defined as “the short-term variations of a digital signal’s significant instants from their ideal positions in time”
- Measured by Eye diagram, Bathtub, ...
- Classified as Random jitter and Deterministic jitter.

Jitter Measurement



Jitter Histogram



Bathtub Curve

Jitter Classification

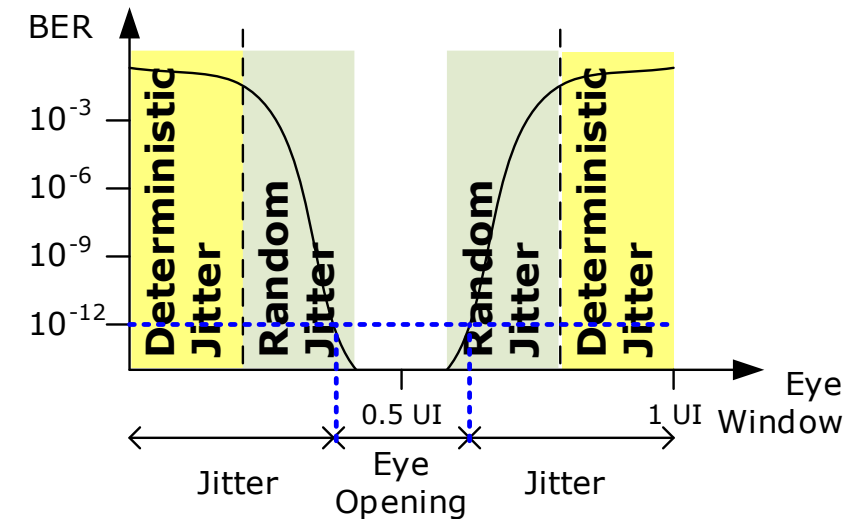
- ✓ Random Jitter*
 - Thermal Noise
 - Shot Noise
 - Flicker Noise
- ✓ Deterministic Jitter**
 - Inter-symbol Interference (ISI)
 - Crosstalk
 - Duty-Cycle Distortion (DCD)
 - Power supply Induced Jitter (PSIJ)

* Gaussian Distribution

** Major topics of DRAM interfaces!!

Jitter and BER Consideration

- What is BER (Bit Error Rate) ?
 - $\text{BER} = (\text{Number of error bits}) / (\text{Number of total transmitted bits})$
 - Target BER of DRAM $\sim 10^{-16} \sim 10^{-12}$
 - Determined by total jitter
 - Smaller jitter is the best way to achieve the target BER.
 - Additional tools to improve BER in DRAM
 - CRC, Link ECC, System ECC, ...



Link Protection : CRC

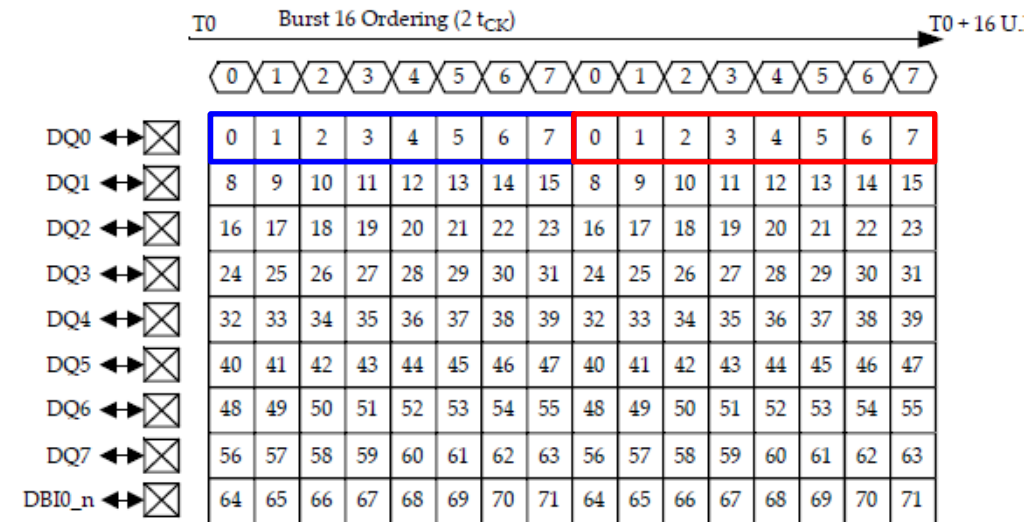
- ❑ CRC ? Cyclic Redundancy Check
- ❑ Error detecting code to detect accidental change in DATA
- ❑ N-bit CRC is additionally transferred with regular data for reliability
 - Cost : Bandwidth overhead (Ex : 16bit CRC + 128 data in GDDR6)
- ❑ Corrective action is taken (Retry), once error is detected.
- ❑ Detection capability of n-bit CRC :
 - All single & double bit errors, any odd number of errors
 - Single error burst not longer than n bits
 - All longer bursts with approximate probability of $(1-2^{-n})$

Link Protection : CRC

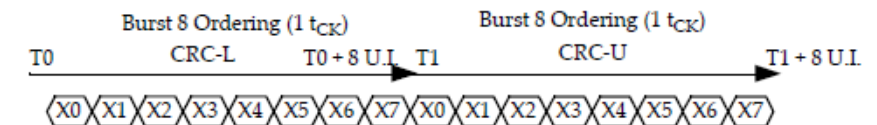
* GDDR6 JEDEC Specification

- ❑ CRC bit ordering
→ Due to the large bus width (up to x512) of DRAM IO, the SI quality of each DQ also has large variation.
- ❑ CRC bit ordering is optimized to detect multiple error in one DQ, because the DQ with poor SI is likely to have multiple errors per transmission.
- ❑ DFE is another source of burst error in one DQ.

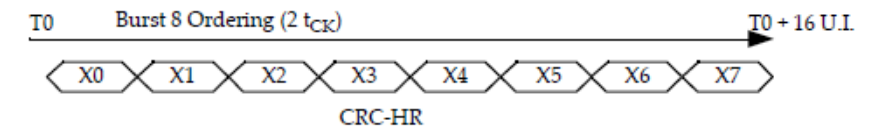
CRC Data Input
DQ/DBI bit ordering



Full data rate
EDC bit ordering

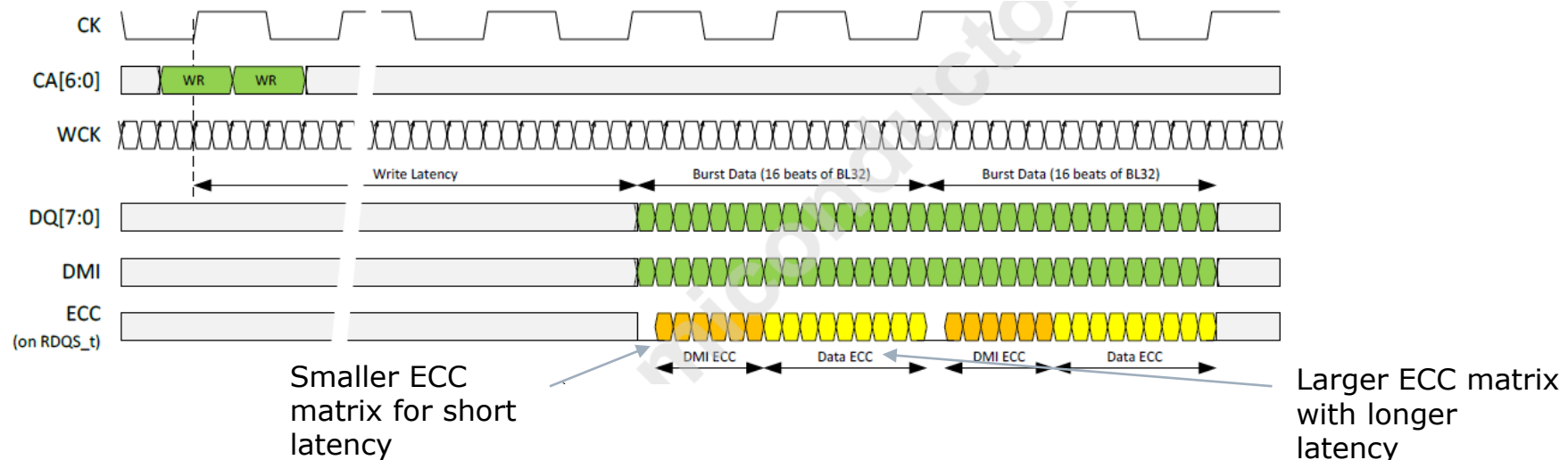


Half data rate
EDC bit ordering

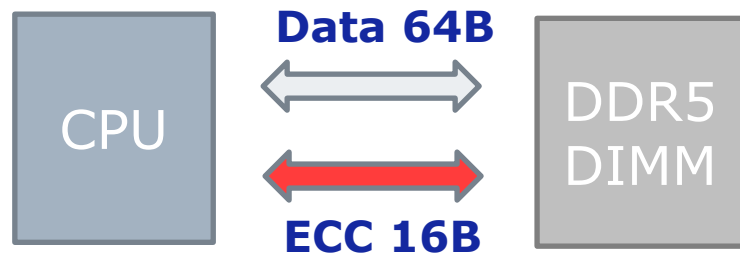


Link Protection : Link ECC

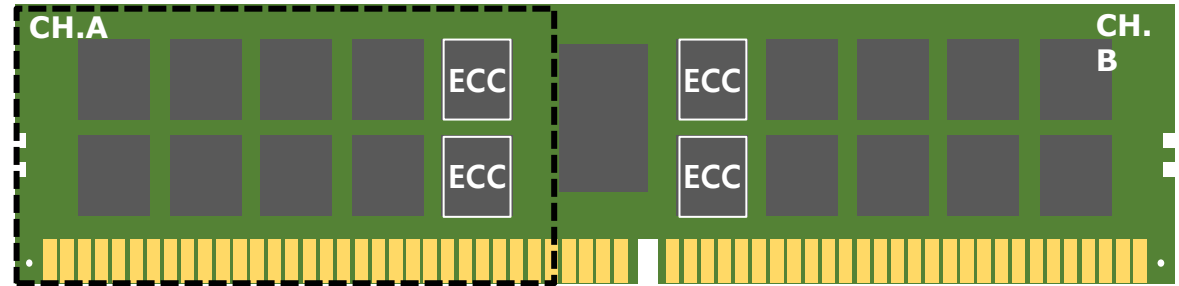
- ❑ Link ECC usually has limited error correction/detection coverage (typically SEC-DEC, single error correction-double error detection)
- ❑ Unlike CRC link protection, retry is not required if single error occurs in DRAM IO interface.
 - Memory controller design is simplified (no retry queue).
- ❑ ECC Encoding/Decoding latency is important for memory performance.



Link Protection : System ECC



DDR5 10x4 module



Access Granularity : Data 64B + ECC 16B

- ❑ Implemented in DDR5/HBM.
- ❑ 10x4 DDR5 RDIMM has dedicated 2 DRAM chips to store system ECC.
- ❑ The system ECC protects system from both DRAM component errors (Cell & Core) and IO interface errors simultaneously.
- ❑ The 16B ECC parity bits are able to cover any failure in a single memory chip.

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DDR5 I/O Margin Validation

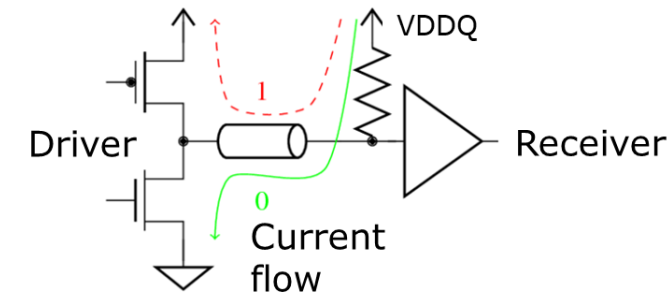


- ❑ Factors must be considered in IO margin validation :
- Process variation of memory controller & DRAM
 - Various PCB channel design (SI/PI of PCB at final user is hardly known)
 - Temperature & Voltage drift during operation
 - Training error depending on training time & algorithm
 - Target BER depending on application

GDDR5 DBI DC (+AC)

□ DBI (data bus inversion) DC/AC?

- DBI DC : In VDDQ terminated interface, driving low level makes the interface consume the termination power.
 - Minimizing the number of DQ's driving low level brings power reduction.
 - By adding 1 DBI pin, the number of DQ's driving low level can be reduced to be same or less than four.
- DBI AC : The number of toggle in the channel is minimized by DBI.
 - As the interface speed increases, the portion of termination power becomes less significant than toggle power.
 - Many IO interface such as HBM does not support the termination, allowing DBI AC feature to reduce the power.

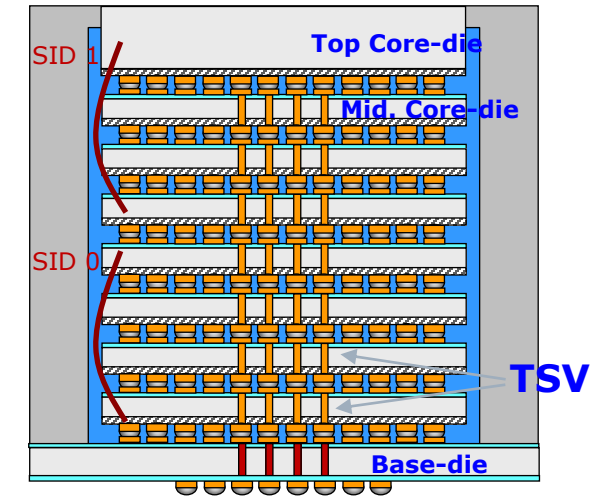


DQ	Raw Data	DBI-DC Encoded
0	L	H
1	L	H
2	L	H
3	H	L
4	L	H
5	L	H
6	L	H
7	H	L
DBI		L

HBM3E/4

HBM (8H) Architecture

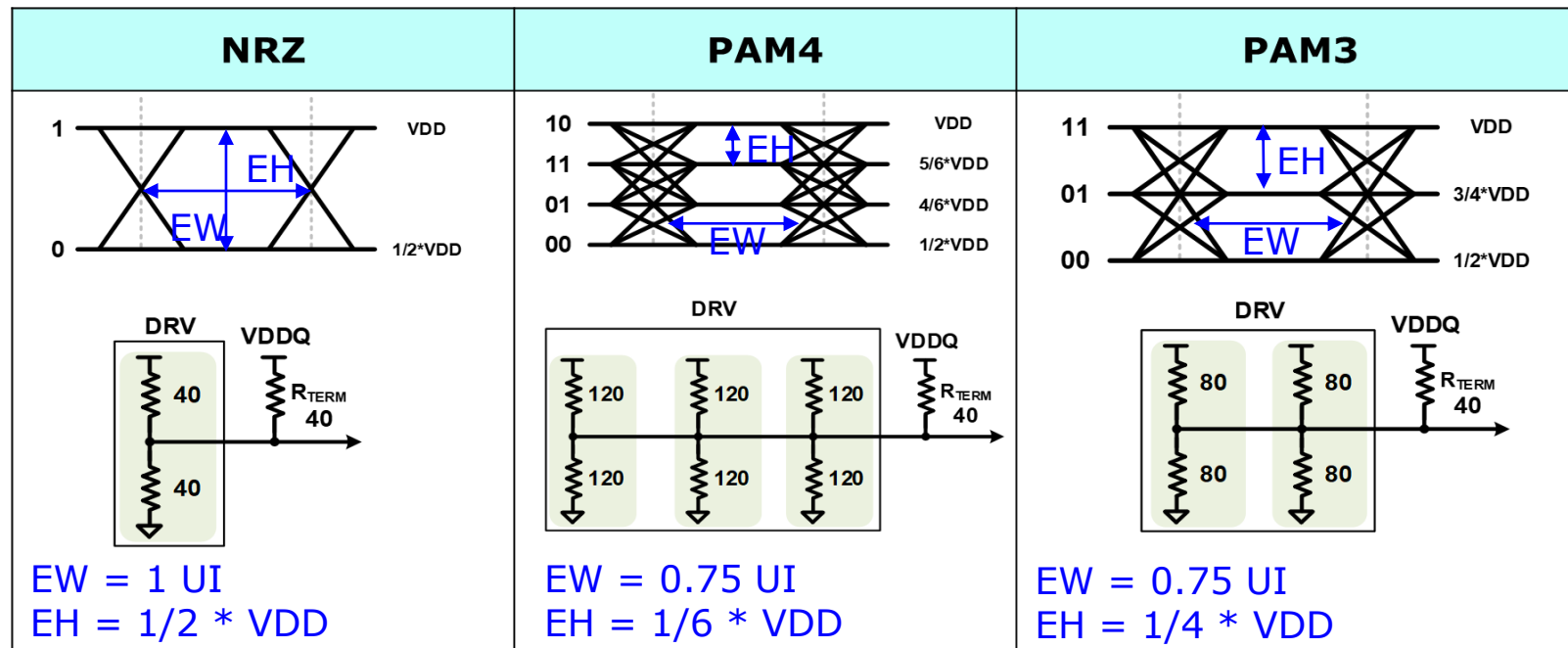
- To Maximize memory bandwidth, design constraints in more than a thousand of IO interface :
 - ODT is not allowed due to too much static power consumption
 - C_{IO} must be minimized for SI.
 - Base die without DRAM cell supports the requirements as buffers.
 - Reduced C_{IO}
 - Better Tr. Performance
 - Simpler IO design
 - IO shared between core dies.
 - Due to above constraints, IO width is increased rather than bit rate even in new generations.



Mode		HBM3e	HBM4
Pin Speed		8.0Gbps~10Gbps	8.0Gbps~10Gbps
# of DQ		1024	2048
Max. BW		1.0TB	2.0TB
Power	VDDQ	1.1V	0.7~0.9V
	VDDQL	0.4V	0.4V
	VDDC	1.1V	1.0~1.05V
	VPPE	1.8V	1.8V

GDDR7 PAM3

- GDDR7 is the first JEDEC Standard DRAM adopting PAM signaling (GDDR6x was not JEDEC standard, the custom DRAM).
- With the same clock frequency, the data rate of PAM3 or 4 will be $\sim 2x$ compared to that of NRZ.

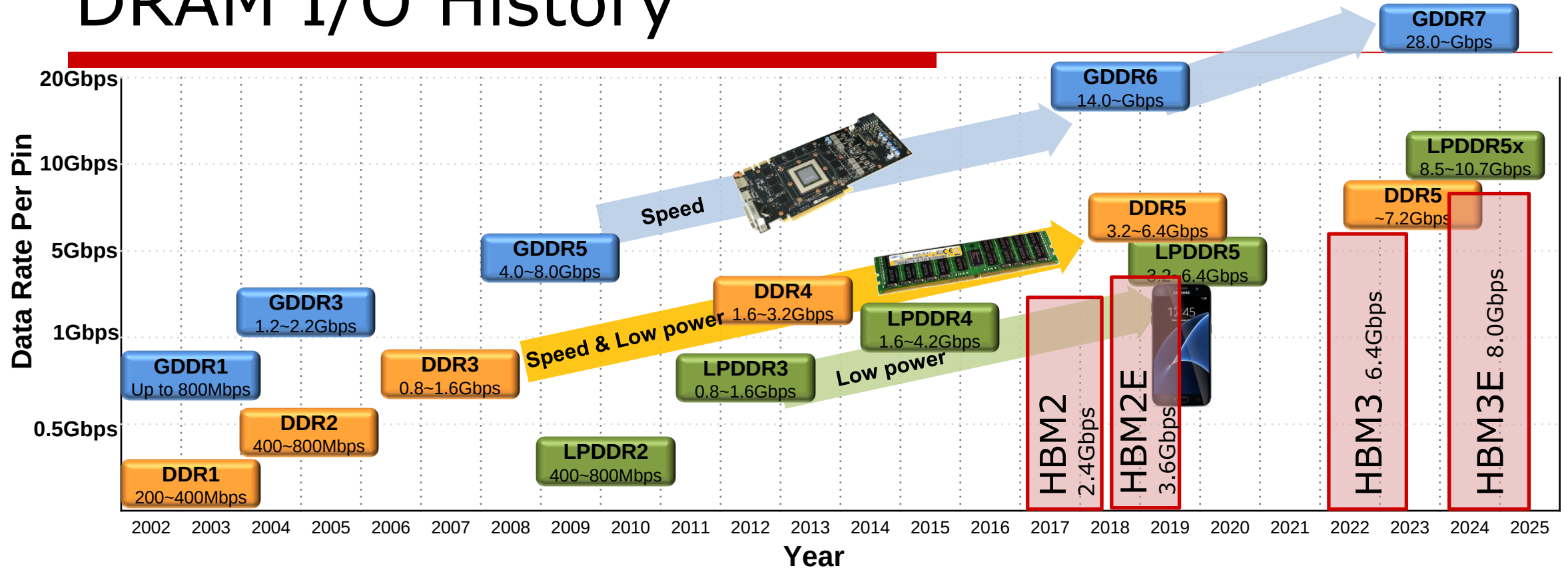


* Assumption : TX impedance = GPU termination = 40 Ohm

LPDDR6 IO

- Increased bus width (X16 → X24) without doubling bit-rate?
Regarding the constraints in lower power interface (LPDDR_x) ←
Power Efficiency!!
- DRAM transistors are slower than SoC transistors for now.
 - Cell capacitance fabrication process requires high temperature.
→ Transistor doping profile changes, degrading performance.
 - Metal lines are located above cell capacitance while transistors are located below.
→ Large parasitic capacitance unavoidable.
- Multi-tap DFE is undesirable due to its power consumption.
- Backward compatibility (LP5x/LP6 combo controller) also prohibit LPDDR_x from adopting radical interface change such as PAM.

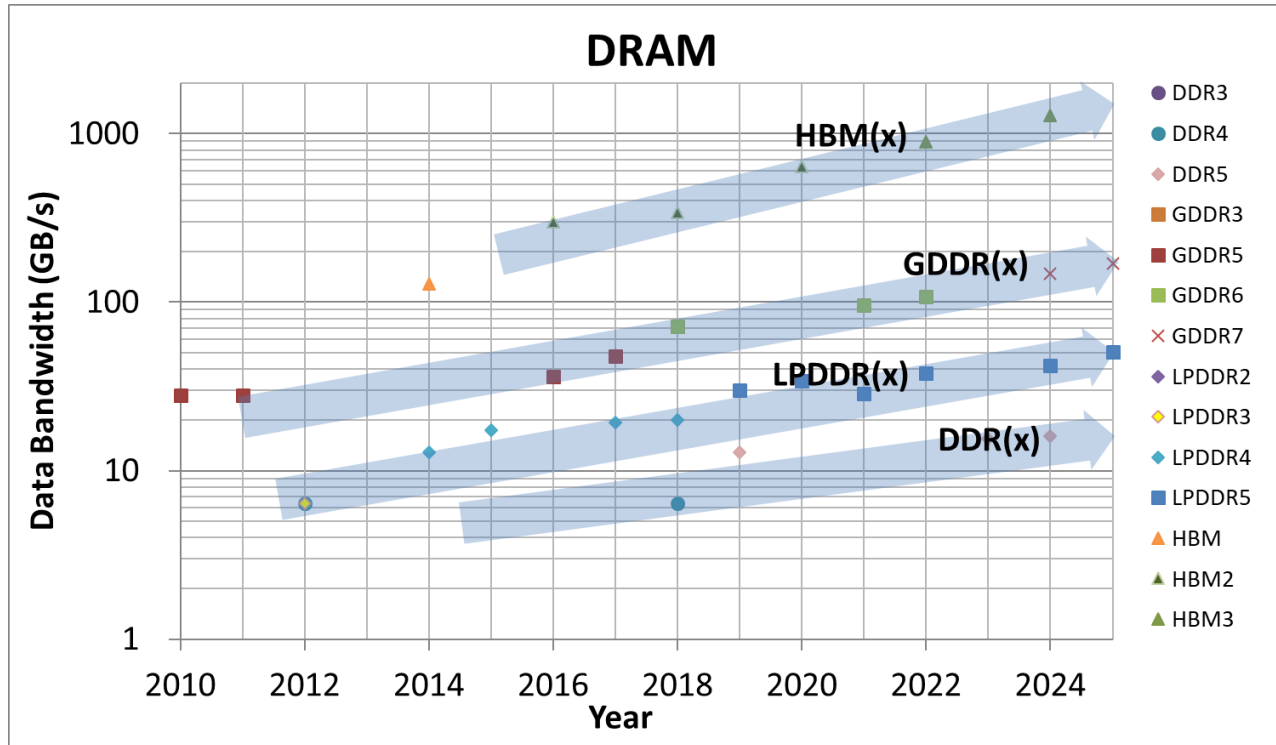
DRAM I/O History



- Interface speed has been doubled in every 3~5 years.
- Historically, GDDRx always offered highest per pin bandwidth by adopting new IO technology first. (VDDQ termination, DBI-DC, CRC, WCK clocking, HKMG process, PAM3)
 - * GDDR3
 - * GDDR5
 - * GDDR6
 - * GDDR7

DRAM Trend in the Future

□ How to increase Data Bandwidth?



1. Increasing per-pin speed

- Cost-effective
- Multi-level signaling : PAM3, PAM4
- GDDR7 in ISSCC2024

2. Increasing the number of data pins

- Simple way but high cost
- HBM4

➔ *What solution will be preferred for DDR and LPDDR?*

DRAM I/O in ISSCC2025

Memory interface

- 22.1 A 0.275pJ/b 42Gb/s/pin Clock-Referenced [PAM3 Transceiver](#) Tolerant to Supply noise, Reference offset and Crosstalk for Chiplets and Short-reach Memory Interfaces.
- 22.2 An 850μW, 2-to-5GHz Jitter-Filtering and Instant-Toggling Injection-Locked [Quadrature-Clock Generator](#) for Low-Power Clock Distribution in HBM Interfaces
- 22.3 A 42-Gbps [Single-Ended PAM-3 Receiver](#) with Hybrid DFE for GDDR7 Memory Interfaces
- 22.4 A 32-50Gb/s/pin [Single-Ended PAM-4 Transmitter](#) with ZQ-Based FFE and PAM-4 LSB DBI-DC Encoding
- 22.5 A 0.3pJ/b 32Gb/s/pin [Single-Ended PAM-4 Receiver](#) with Delay-less Capacitive Feedback Equalizer

Nonvolatile memory and DRAM

- 30.3 A 24Gb 42.5Gb/s GDDR7 DRAM with low-power [WCK distribution, R/C optimized dual-emphasis TX](#), and voltage/time margin enhancement with power reduction.
- 30.4 A 16Gb 12.7Gb/s/pin LPDDR5 Ultra Pro DRAM with Novel [4-Phase Self-Calibration and AC-Coupled Transceiver Equalization](#) in the Fifth Generation 10nm DRAM Process.

Conclusion

- DRAM I/O for various applications is Introduced:
 - Application for DDR, LPDDR, GDDR, and HBM
 - Signaling and clocking
- Techniques to extend I/O speed are Reviewed:
 - Channel and Equalization
 - Training and Link Protection
- Present and Future DRAM :
 - DDR5, GDDR7, LPDDR6, HBM3/4.
- Two tracks to increase data bandwidth of DRAM:
 - Increase the number of pins like HBM, or the per-pin speed with multi-level signaling like PAM3 in GDDR7

References

1. JESD79-4, JEDEC DDR4 SDRAM specification
2. JESD79-5, JEDEC DDR5 SDRAM specification
3. JESD250, JEDEC GDDR6 specification
4. JESD235, JEDEC HBM specification
5. JESD209-5, JEDEC LPDDR5 specification
6. JESD239, JEDEC GDDR7 specification
7. Design of Analog CMOS Integrated Circuits, Behzad Razavi, McGRAW-HILL
8. Digital System Engineering, Dally and Poulton, Cambridge